SETUP AND SETTINGS · 10 INCH · FIRST EDITION

Samba Solar Track



From the first switch-on to a fully set-up tracker: setup at the display, setup program, location and sun, all adjustable values.

Who is it for?
For everyone setting up the controller for the first time, even without any prior knowledge. Ready-made recommendations for the Samba Solar Rack, and for any other system a guide to finding the right value.

One or two axes
Both operating modes are described equally. Anything that applies only to one axis carries the label **1 axis**, anything that applies only to two axes the label **2 axes**.

Status
First edition, November 2026, firmware 0.7.0. The firmware has only been tested on the PC and in the simulation, not on a real rack.

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Samba Solar Track

Setup and settings · 10-inch version

DE | **EN** | FR | ES | NL | PT | PL

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First edition: firmware 0.7.0 tested on the PC only

Everything this manual describes is implemented in firmware 0.7.0 and checked on the PC and in the simulation (automatic tests, images from the real user interface code). It has **not yet been tried** on a real rack with drives, anemometer and sun sensor; the RearCircuitBoard_10 board has not been manufactured yet. Limits for the checks of the setup program will only be set once the drives have been delivered and measured on the rack. Assumptions are marked in the text with Assumption.

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1 About this manual

This manual guides you from the first switch-on of the controller to a fully set-up tracker. You need no prior knowledge. Every technical term is in the glossary.

What this manual covers

- The **setup at the display**: language, operating mode, anemometer, network, time, remote access, design, sound, screen and firmware (Chapter 4).
- The **setup program**: on the finished rack it checks every function and every travel path step by step. Only after that does the system track by itself (Chapter 5).
- **Location, orientation, sun and night**: where the rack stands, which way it faces, when it is night (Chapter 6).
- **Every adjustable value** with its meaning, default, limits, a recommendation for the Samba Solar Rack and a guide for any other system (Chapter 7).
- What you can set **in the browser** and what only works at the display (Chapter 8).

What it does not cover

- Assembling the mechanics. For that, see the **assembly manual of the Samba Solar Rack** (website and shop) or the instructions for your frame.
- Connecting the cables. For that, see the wiring manual (German only; file `verkabelung.html` in the same folder).
- The circuit boards, flashing the firmware and checking the hardware. For that, see the manual for assembly and operation (file `index.html`).

> ⚠️ **Everything must be connected first**
> Only start with this manual once the rack is assembled, the wiring is finished and checked (wiring manual, German only) and the controller has power. The setup program moves the rack.

One or two axes: how the sections are marked

Samba Solar Track works in two operating modes. Most users have **two axes**; both operating modes are described equally here. Anything that does not apply to both carries a label:

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↕

1 axis

Operating mode “vertical only”. The rack only changes its tilt (elevation). This is how the Samba Solar Rack works.

⟲

2 axes

Operating mode “vertical + horizontal”. The rack tilts and also turns from east to west (azimuth).

✓

both

Applies to every system. Sections without a label also apply to both.

○

optional

May be skipped. The controller also works safely without this item.

How to read the steps

Every step in the setup program has four fields:

WHAT YOU SEE
What the display shows. Almost always with a picture of the screen.

WHAT YOU DO
Which button you press and what you do at the rack.

WHAT MUST HAPPEN
The expected result. Only continue when it is right.

IF NOT
What to do then. More help is in troubleshooting.

ℹ️ **The display pictures**
All screen pictures come from the real user interface code, generated on the PC. On the device, numbers (wind, brightness, time) may look different. The design shown is “Standard” in English. Pictures with 1 axis are called `y1_…`, pictures with 2 axes `y2_…`.

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2 The way to a finished system

The controller is set up in four stages. The order is fixed: without setup no drive moves, without the setup program the system does not track by itself.

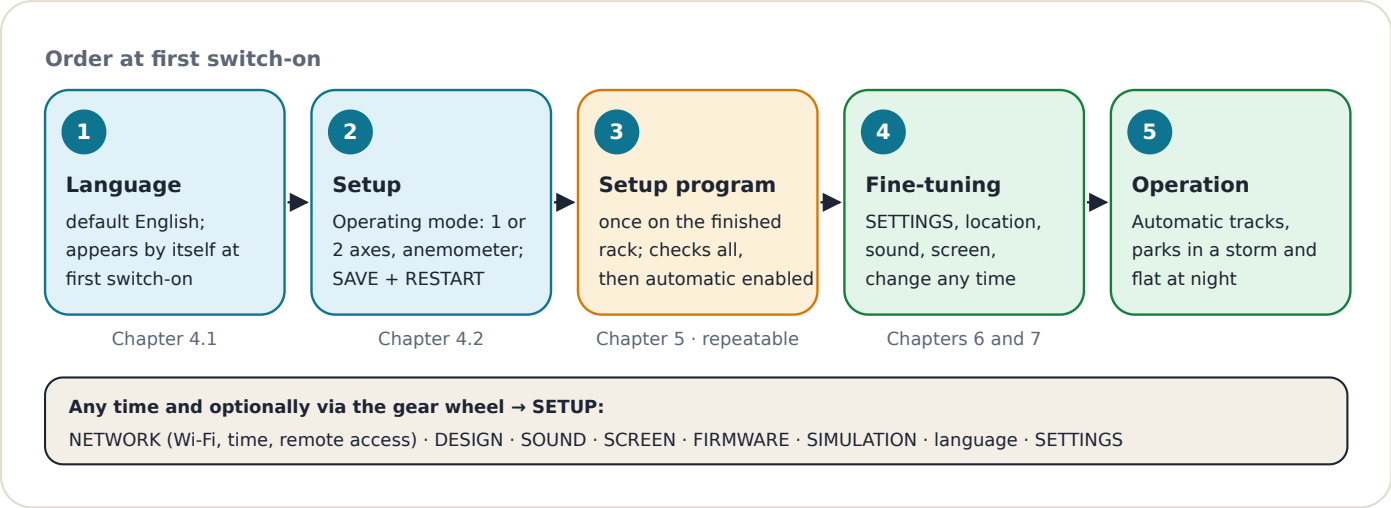


Fig. 1 – The four stages of the setup. You can do network, design, sound and screen at any time later.

1 Choose the language

At first switch-on the language selection appears by itself, set to English by default. The choice applies immediately. *Result:* All texts are in your language. Chapter 4.1

2 Setup: operating mode and anemometer

You choose 1 or 2 axes and your anemometer and tap SAVE + RESTART. *Result:* The controller restarts. From now on storm protection and anemometer monitoring are active. Chapter 4.2

3 Setup program

It opens by itself after the restart and checks everything on the rack one after the other: board, anemometer, travel directions, travel paths, sun sensor, test run. *Result:* “Done! All checks are passed.” The automatic is enabled. Chapter 5

4 Fine-tuning

Location and orientation, tracking values, sound, screen, remote access. *Result:* The system works the way you want. Chapter 6 and 7

💡 **How long does it take?**

The setup takes a few minutes. Take your time for the setup program and choose a day with little wind and some sun: the drives travel the whole way several times and need pauses in between (duty cycle, Chapter 7). You can interrupt the setup program at any time with CONTINUE LATER and resume later at the same point.

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What the controller does at each stage

STAGE	DRIVES	STORM PROTECTION	AUTOMATIC (TRACKING, NIGHT)
Setup missing	never move	not yet	off
Setup saved, setup program open	only in the setup program, in manual mode and in a storm	**active**: in a storm the rack moves flat	locked, purple band “SETUP PROGRAM MISSING”
Setup program completed	normal	active	**enabled**

⚠️ **The direction of the flat position is only checked in step 5**

Storm protection and parking move the rack “flat”. Which direction that is on your rack is checked by the setup program in step 5. Until then the default applies. So do not start the setup program in a storm, and stay with it until step 5 is passed.

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3 Axes and directions

Before the rack moves for the first time: these words come up again and again on the display. They apply to every rack.

One axis: the tilt (elevation) 1 axis also with 2 axes

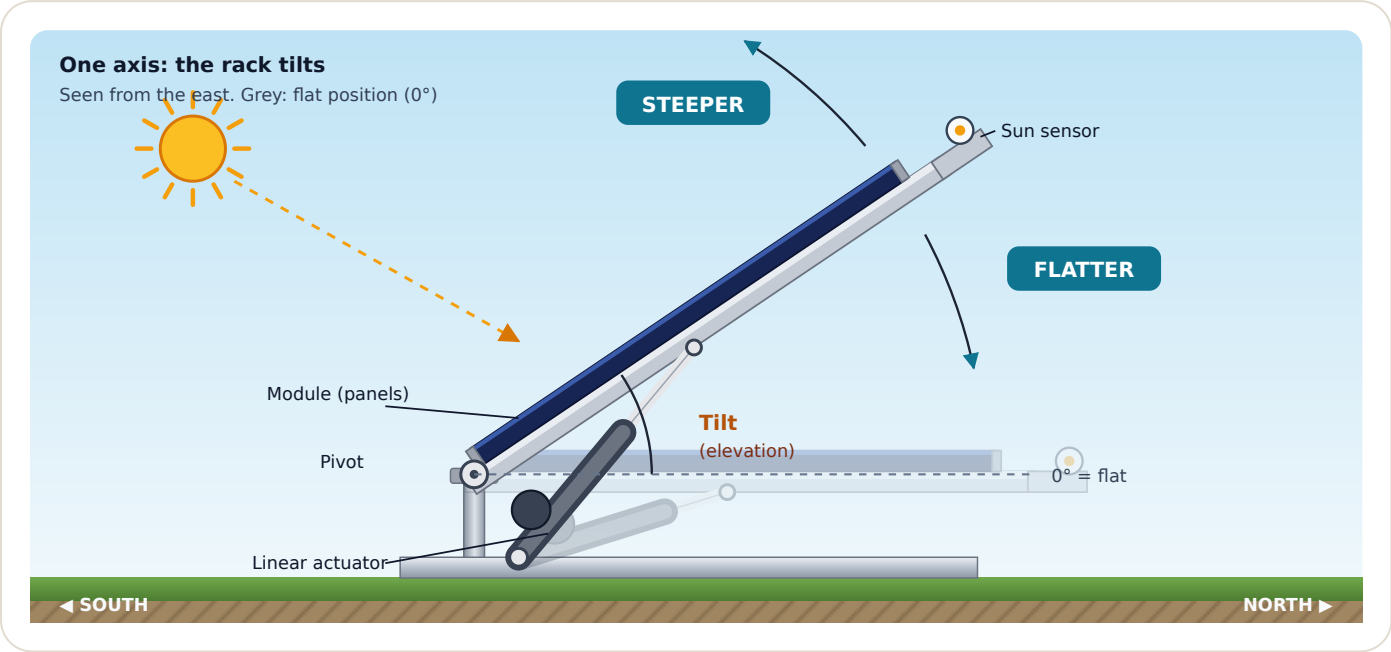


Fig. 2 – A rack with one axis from the side (example, simplified). The drive lifts the high end of the module. STEEPER raises the panels, FLATTER lays them down. The Samba Solar Rack tilts between 0° and 60°. Your frame may be built differently.

- **STEEPER**: the panels rise, their top edge goes up. Needed in the morning, in the evening and in winter, when the sun is low.
- **FLATTER**: the panels lie down, towards horizontal. Needed at midday in summer, in a storm and at night.
- The buttons on the display are called STEEPER and FLATTER. The terminals of the board are still called UP and DOWN. Which terminal means “flatter” is set by the value “Flat position is in UP direction”; the setup program checks it.

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Two axes: tilt and rotation (azimuth) 2 axes

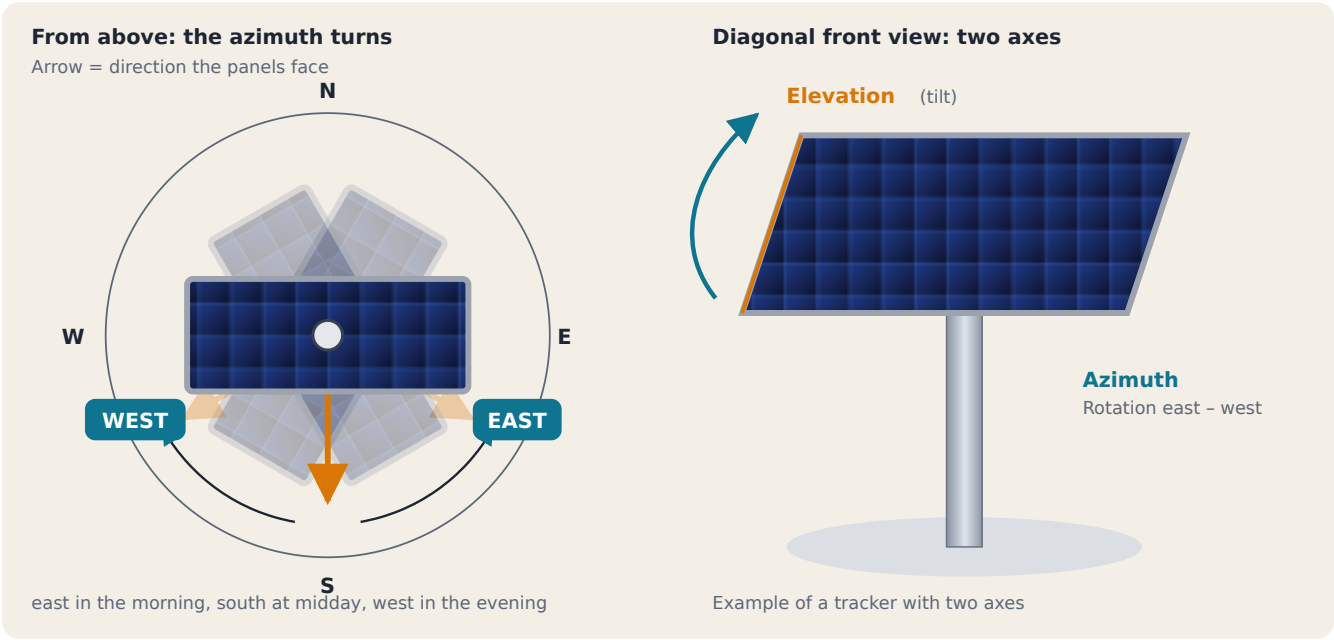


Fig. 3 – A tracker with two axes (example). Left, from above: the azimuth turns the panels east in the morning, south at midday, west in the evening. Right: the elevation tilts them as with one axis.

- **EAST**: the rack turns towards the morning sun. On the board output this is the EAST contact (terminal J5).
- **WEST**: the rack turns towards the evening sun (WEST contact, terminal J6).
- Only one axis moves at a time. This spares the power supply.
- The Samba Solar Rack only has the tilt. Anyone with two axes uses a different frame; for that, every value in Chapter 7 says how to find the right value.

Flat position: the safe position

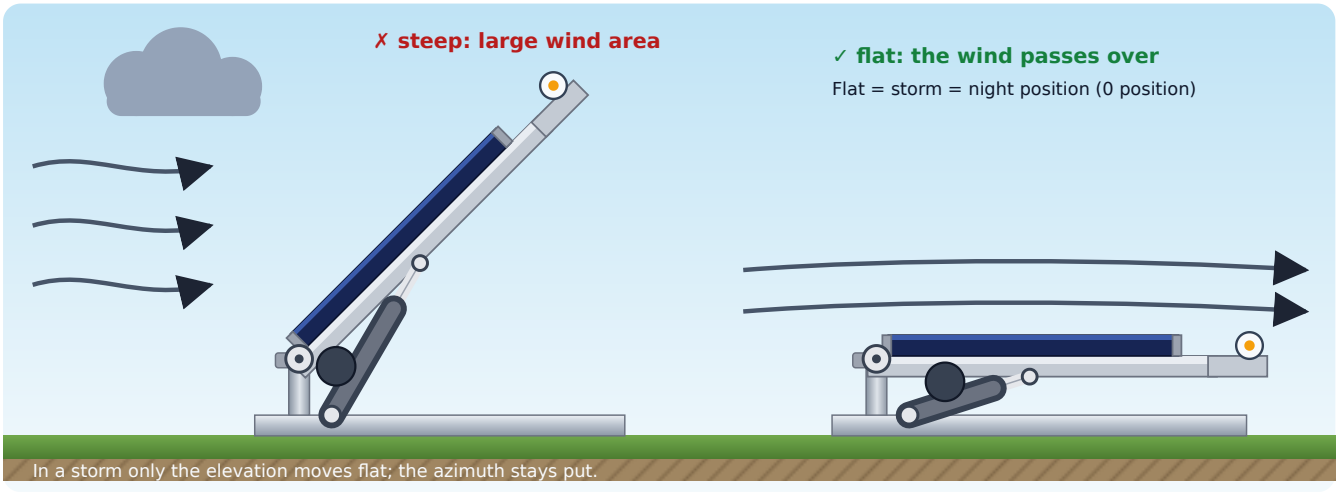


Fig. 4 – Flat, a module offers the wind the smallest area. That is why the controller moves to the flat position in a storm, on anemometer failure, when parking and at night.

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CAUSE	ELEVATION	AZIMUTH 2 AXES	PRIORITY
Storm (gust above the storm threshold)	moves flat, stays flat until the hold time ends	stays put	highest; cannot be overridden
Anemometer not responding	moves flat, automatic locked	stays put	high; manual mode remains possible, but then without storm protection
PARK FLAT (button)	moves flat, stays until CANCEL PARKING	stays put	above the automatic
Night	moves flat (0 position)	then east, if “Return east at night” is on	normal move, no priority move

ℹ️ **0 position = flat position**

The firmware calls the night position the “0 position”. It lies in the same direction as the storm position (decided during implementation, Assumption: with every drive variant, the end position in the flat direction is also the sensible night position).

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4 Setup at the display

The setup appears by itself at the first switch-on. Later you reach it via the gear wheel at the top right of the dashboard. If the device is already set up, the controller including storm protection keeps working while the setup is open.

⛔ **Without setup nothing moves, and there is no storm protection**
As long as operating mode and anemometer are not saved, all outputs stay off. The BEREIT (ready) indicator light flashes. So set up the controller right after the first switch-on.

4.1 Language both

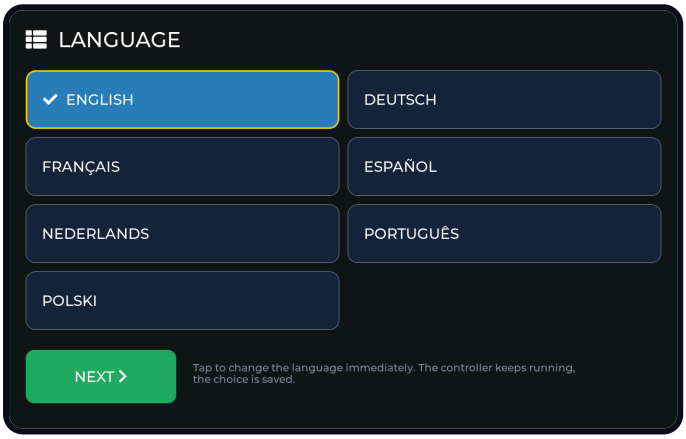


Fig. 5 – The language selection. At the first switch-on it is set to ENGLISH.

1 **Tap a language**
The choices are English, Deutsch, Français, Español, Nederlands, Português and Polski.
Result: The texts change immediately, without a restart. The choice is saved.

2 **Tap NEXT**
Result: The setup appears.

Later you change the language with the button showing the name of the language at the top right of the setup (for example “ENGLISH …”). The web interface always shows the language of the display.

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4.2 Operating mode and anemometer both

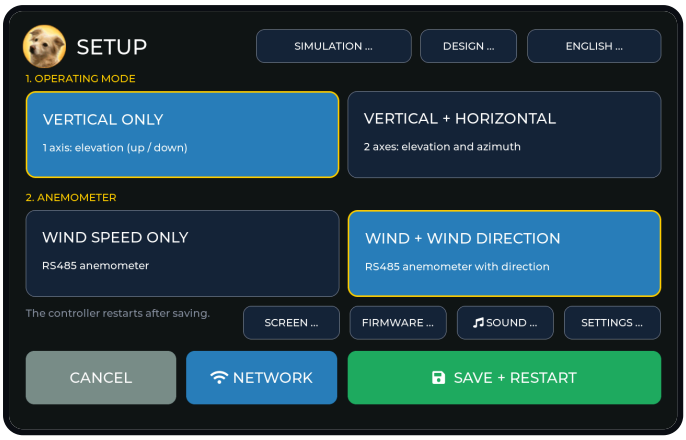


Fig. 6 – The setup. At the top the operating mode, below it the anemometer, at the bottom SAVE + RESTART. Selected here: vertical only and wind + wind direction.

1

Choose the operating mode

VERTICAL ONLY (1 axis: elevation) or VERTICAL + HORIZONTAL (2 axes: elevation and azimuth). Nothing is selected by default. *Result:* The selected button is blue with a yellow border.

2

Choose the anemometer

WIND SPEED ONLY or WIND + WIND DIRECTION. Both are anemometers with an RS485 output (Modbus RTU); other outputs (pulse, 0–10 V, 4–20 mA) are not supported. *Result:* The button is marked.

3

Tap SAVE + RESTART

The button only works once both are selected. *Result:* All outputs switch off, the controller restarts, then the setup program opens.

1 axis vertical only

The azimuth outputs always stay open. The display has no EAST and WEST buttons; instead there is the TILT card with fixed angles (0°, 30°, 60° and two preset angles). Of the sun sensor, only north versus south counts.

2 axes vertical + horizontal

Both axes are tracked, always one after the other. The display has STEEPER, FLATTER, EAST and WEST. The sun sensor is evaluated on all four sides. There is no TILT card with fixed angles.

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ℹ️ **Which anemometer?**
For the Samba Solar Rack, two separate sensors on the same bus are intended: wind speed (address 3) and wind direction (address 4), i.e. WIND + WIND DIRECTION. The wind direction is only displayed and logged; it affects neither tracking nor storm protection. Anyone with only a speed sensor chooses WIND SPEED ONLY.

⚠️ **Changing later**
You can change operating mode and anemometer at any time. New settings only apply after SAVE + RESTART; until then the controller keeps working with the previous ones. CANCEL discards the selection. If you have changed the operating mode, check the system again afterwards: SETUP PROGRAM → CHECK EVERYTHING AGAIN (Chapter 5.4).

4.3 The other buttons of the setup

BUTTON	PURPOSE	APPLIES	MANDATORY?
NETWORK	Wi-Fi, hotspot, time, remote access (4.4)	after SAVE + RESTART	optional, needed for the time
DESIGN …	Colours of the user interface (4.6)	immediately	optional
SOUND …	Confirmation tones, volume (4.7)	immediately	optional
SCREEN …	Idle mode, motion sensor, battery (4.8)	immediately	optional
FIRMWARE …	check for and install new firmware (4.9)	after confirmation	optional
SETTINGS …	all adjustable values, setup program (Chapter 7); only once the setup is saved	differs per value	–
SIMULATION …	Testing without the RearCircuitBoard_10 board (4.10)	after confirmation and restart	only for trying out
ENGLISH … (name of the language)	Language selection (4.1)	immediately	–

4.4 Network: Wi-Fi, hotspot and time optional

Wi-Fi is off by default. The controller never needs it: tracking and storm protection work just the same without a network. With Wi-Fi, however, you get the **exact time** (then the device knows sunrise and sunset), the **web interface** for phone and PC, and the **check for new firmware**.

ℹ️ **No Wi-Fi, no time**
The board has no clock of its own with battery backup. The time only comes over Wi-Fi from a time server. Without Wi-Fi the device detects night by the brightness (Chapter 6).

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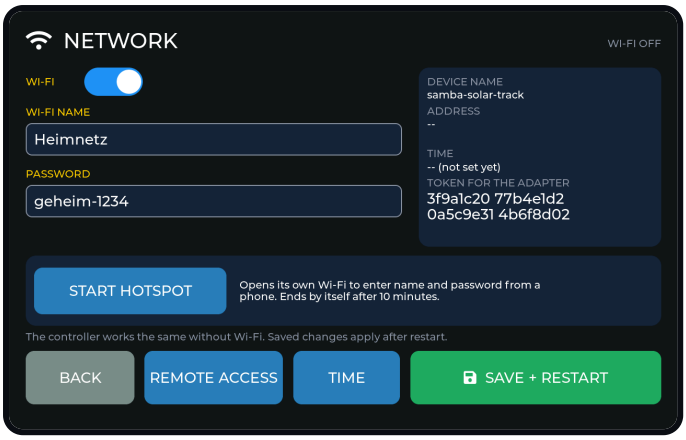


Fig. 7 – NETWORK: Wi-Fi switch, Wi-Fi name, password, on the right device name, address and token.

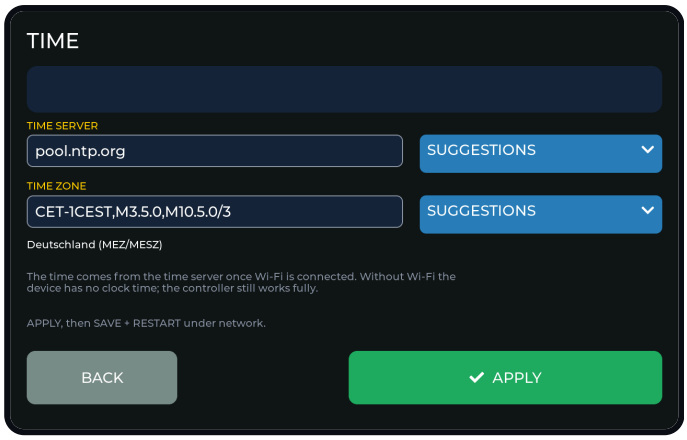


Fig. 8 – TIME: time server and time zone, enter freely or choose from SUGGESTIONS.

Way A: type it in at the display

1. ### Setup → NETWORK
   Result: The NETWORK screen appears, “WI-FI OFF” at the top right.

2. ### Switch on Wi-Fi
   Tap the WI-FI switch. *Result:* The switch is blue.

3. ### Enter the Wi-Fi name and password
   Tap the field, type with the on-screen keyboard, confirm with the tick. The name has 1 to 32 characters, the password is empty or has 8 to 63 characters. The keyboard has no umlauts or accents; for such names use way B. *Result:* Name and password are in the fields.

4. ### Check the time
   TIME button. Default: time server `pool.ntp.org`, time zone Germany (CET/CEST). If you live in a different time zone, choose it under SUGGESTIONS and tap APPLY. *Result:* You are back on the NETWORK screen.

5. ### SAVE + RESTART
   Result: All outputs switch off, the device restarts and connects. The Wi-Fi symbol appears in the header: white = connected, yellow = no connection. Shortly afterwards the time appears to its left.

Way B: via the setup hotspot (with your phone)

1. ### NETWORK → START HOTSPOT
   Result: The device opens its own Wi-Fi “Samba-Setup-” with four characters. Name and password are shown on the display; the password is different at every start.

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2

Connect your phone to this Wi-Fi and open `http://192.168.4.1` in the browser

Result: The setup page appears (picture on the right) with a list of the Wi-Fi networks found.

3

Choose the Wi-Fi, enter the password and device name, save

The device name consists of a–z, 0–9 and hyphen, by default `samba-solar-track`. *Result:* The device restarts after 1.5 s and connects to your Wi-Fi.

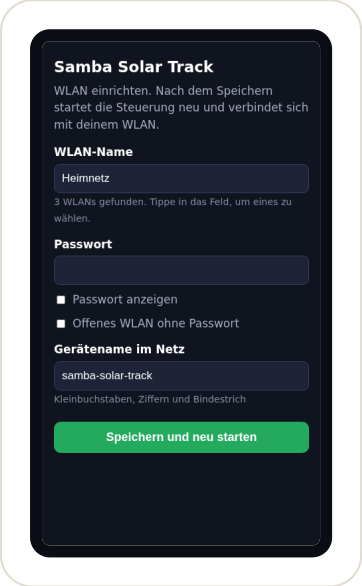


Fig. 9 – The setup page of the hotspot on the phone.

The hotspot ends by itself after 10 minutes. While it is running, the Wi-Fi connection rests.

💡 **Where do I find the device on the network?**

At the address shown on the NETWORK screen, or at `http://samba-solar-track.local` (or the device name you have given it). If the connection fails, the device tries again every 30 seconds, without end.

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4.5 Remote access: log in your phone, lock remote control

optional

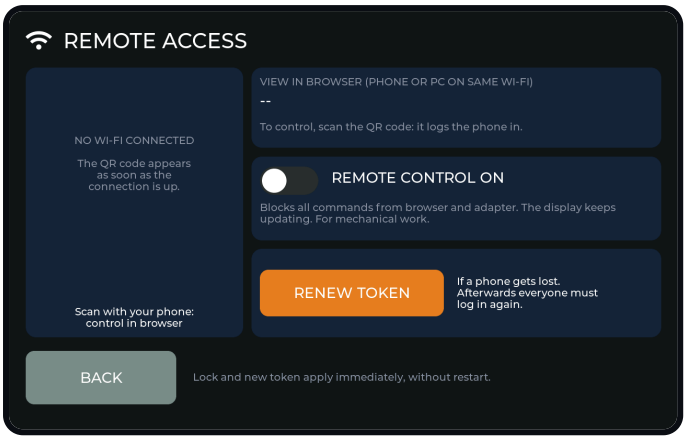


Fig. 10 – REMOTE ACCESS (NETWORK → REMOTE ACCESS): QR code for the phone, REMOTE CONTROL switch, RENEW TOKEN.

- **QR code**: appears as soon as Wi-Fi is connected. Scan it with your phone; the browser opens the web interface and is logged in. Without logging in, the browser can only watch.
- **Token**: a secret key of 32 characters that the device creates itself. On the PC you enter it in the web interface under ENTER TOKEN; the NETWORK screen shows it in four blocks.
- **REMOTE CONTROL**: when locked, the device accepts no remote command and no change; displays keep updating. Lock it when someone is working on the mechanics. Applies immediately.
- **RENEW TOKEN**: if a phone has been lost. With a confirmation prompt; afterwards everyone has to log in again. Applies immediately.

4.6 Design optional

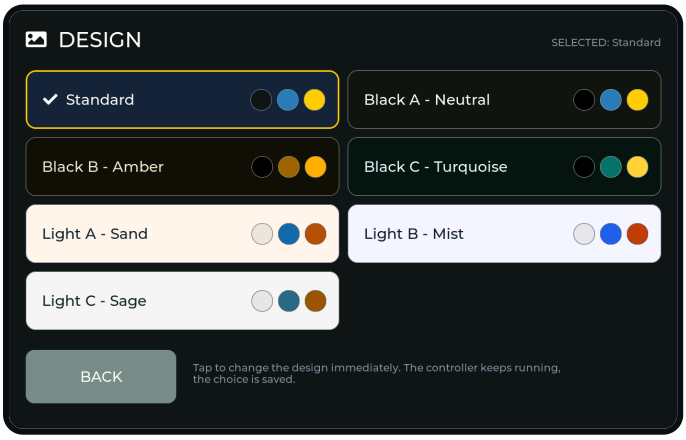


Fig. 11 – Seven designs: Standard, three black ones (for dark rooms), three light ones (for bright rooms).

Tapping changes the design immediately, on the display and in the web interface. The controller keeps running meanwhile.

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4.7 Sound optional

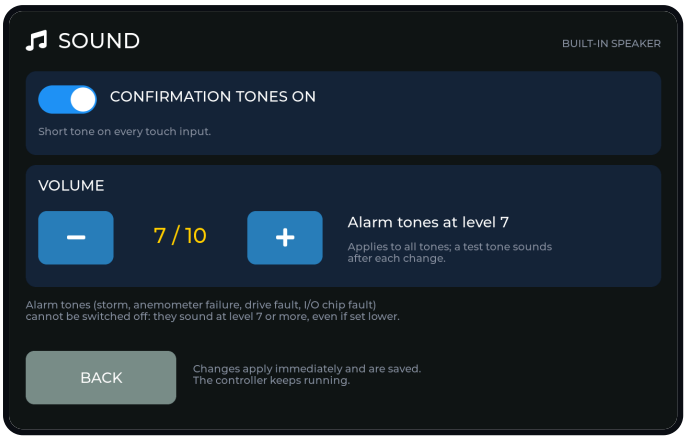


Fig. 12 – SOUND: confirmation tones on or off, volume in ten levels.

SETTING	DEFAULT	POSSIBLE	NOTE
Confirmation tones	on	on / off	short tone on every touch input at the display
Volume	7 of 10	1 to 10	a test tone sounds after every change
Alarm tones	on	only quieter, never off	for storm, anemometer failure, drive fault and fault of a chip; they always sound at level 7 or more

Changes apply immediately and are saved. Sound can only be set at the display.

4.8 Screen: idle mode, motion sensor, battery optional

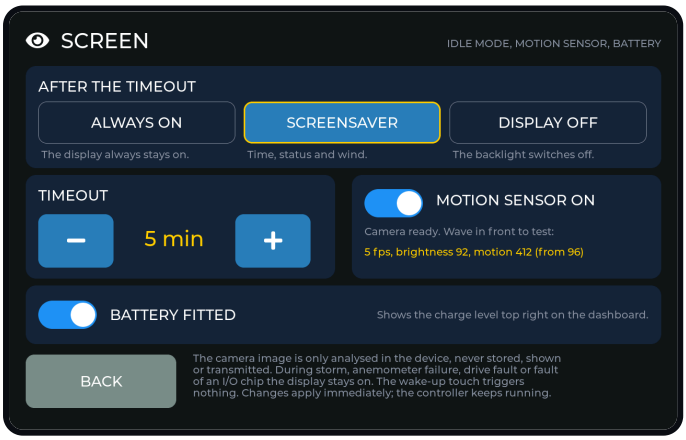


Fig. 13 – SCREEN: what happens after the timeout, timeout, motion sensor, battery fitted.

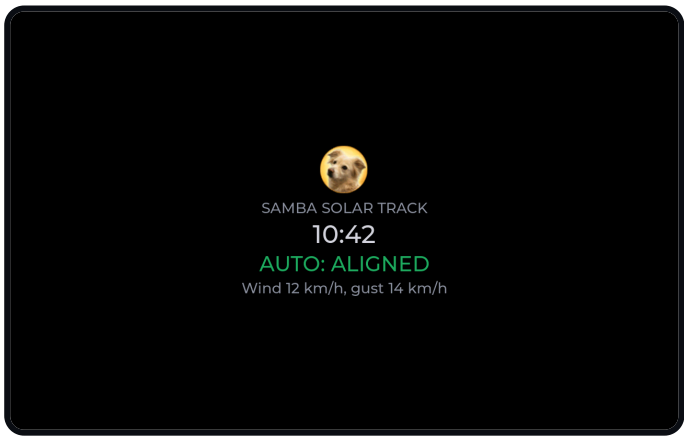


Fig. 14 – The screensaver shows time, status and wind.

SETTING	DEFAULT	POSSIBLE
After the timeout	SCREENSAVER	ALWAYS ON · SCREENSAVER · DISPLAY OFF
Waiting time	5 min	1 to 60 min (buttons − and +)
Motion sensor	on	on / off
Battery fitted	yes	yes / no (shows the charge level at the top right of the dashboard)

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Motion in front of the display or a touch wakes it; the wake-up touch does not trigger any action. In a storm, on anemometer failure, drive fault and fault of a chip, the display switches on by itself. The image of the built-in camera is only analysed in the device, never stored, shown or transmitted.

> 💡 **Checking the motion sensor**
> On the SCREEN screen there is a line with “motion …” below the switch. Wave in front of the display: the number must rise. If it is too dark, it says “too dark”.

4.9 Firmware optional

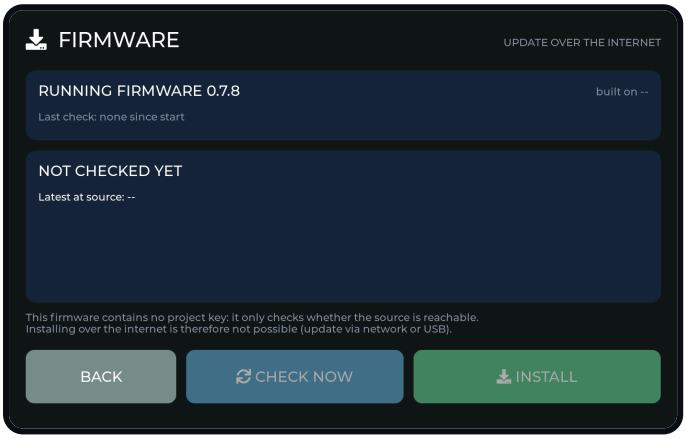


Fig. 15 – FIRMWARE: running firmware, result of the last check, CHECK NOW and INSTALL.

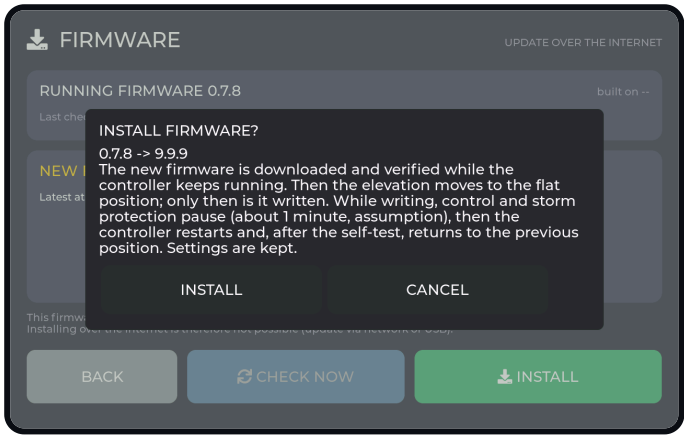


Fig. 16 – The confirmation prompt before installing. The settings are kept.

With Wi-Fi the device checks once a day whether there is newer firmware. It never installs by itself, only when you tap INSTALL and confirm the prompt. Before writing, the elevation moves flat; while writing, control and storm protection pause briefly. Then the device restarts and checks itself; if that fails, it returns to the previous firmware. The saved settings are kept.

4.10 Simulation: trying out without the board

With the simulation you can try out operation without the RearCircuitBoard_10 board, without drives and without sensors, for example at your desk. It can only be switched on when the board is **not** plugged in, and it needs a confirmation. The display shows the simulation permanently. If the board is plugged in at start, a saved simulation switches itself off. The setup program can also be played through in the simulation; its test report is then marked “SIMULATION”.

> ℹ️ **After the simulation the setup program starts again on the real rack**
> A setup program that was only completed in the simulation does not count on the real rack. When the device starts for the first time with the board plugged in, the setup program starts again at step 1 by itself. Until it is passed on the real rack, the automatic stays locked (purple band “SETUP PROGRAM MISSING”); storm protection and parking keep working. You need to do nothing for this except go through the setup program on the rack (Chapter 5).

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5 The setup program

The setup program checks everything on the finished rack once that matters for safe operation: board, anemometer, travel directions, travel paths, sun sensor and the automatic. Where a person has to judge, it asks. Where the device can measure itself, it measures. Only when all mandatory steps are passed does the system track by itself.

> 🧩 **Tested on the PC only**
> The setup program is implemented in firmware 0.7.0 and checked with automatic tests and in the simulation, but has never run on a real rack. There are no limits yet for travel times, Hall steps and angles (decision E-S3): “passed” means end position reached or confirmed by you; the measured values go into the test report.

5.1 Before you begin

- **Little wind.** In a storm every travel test is cancelled, and the rack moves flat.
- **Daylight or a bright lamp.** Sun sensor and test run need light (at least the day threshold, by default sum 6000).
- **Two people are better than one.** One stands at the display, the other watches the rack. If the display is far away, use a phone to talk to each other.
- **A clear view of the rack** and of the anemometer; a ladder for the anemometer if it is mounted high (only climb up the rack with the power supply switched off if you get into the swivel range).
- A **phone with an inclinometer app** (spirit level) for the step “Move to angle” `1 axis` and a **compass** (an app will do) for the location.

How it opens

- After the setup is saved, it opens **by itself**.
- If you leave it (CONTINUE LATER), the dashboard shows the purple band “SETUP PROGRAM MISSING: AUTOMATIC LOCKED – TAP HERE TO CONTINUE” at the bottom. Tapping it opens it again at the same point.
- Also via the gear wheel → SETTINGS → COMMISSIONING → green line **SETUP PROGRAM (CHECK EVERYTHING)**.
- The progress is saved, even across a restart or power failure.

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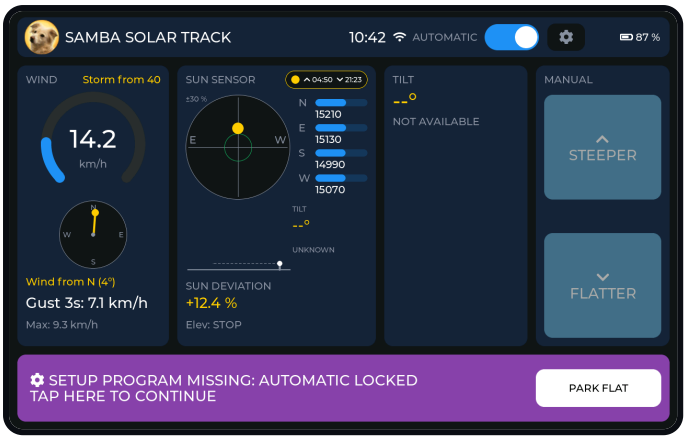


Fig. 17 – As long as the setup program is not completed: purple band at the bottom. PARK FLAT and manual mode still work; storm protection is active.

The buttons at the bottom

BUTTON	WHAT IT DOES
CONTINUE LATER	Leaves the setup program. Every move stops, the progress is kept. The automatic stays locked.
STEP BACK	Goes to the previous step (not in the first step and not after completion).
SKIP	Only for optional steps: Wi-Fi and time, location, lamps/sound/motion sensor. Travel and safety steps can never be skipped.
STOP (red)	Appears during every move. Stops immediately.
REPEAT	After a cancellation or a correction: do the step once more.

⛔ **Every move is a real move**

Limit switches, maximum run time, dead time and duty cycle also apply in the setup program. Still, the drives do not stop if a hand is in the way. Nobody in the swivel range, no animals either. If a storm comes, the step is cancelled and the rack moves flat (picture below).

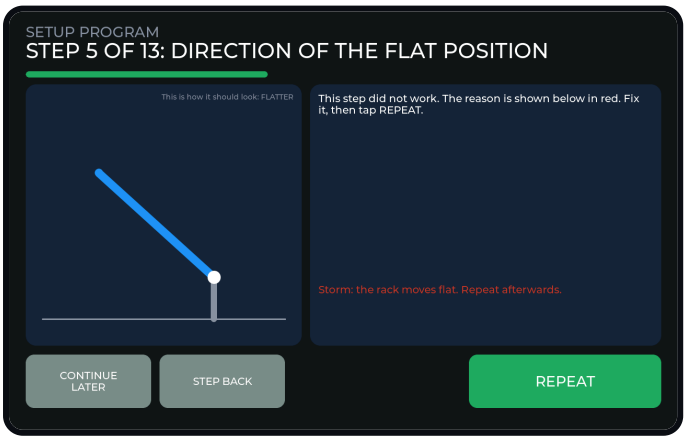


Fig. 18 – Cancelled by a storm, here in step 5 “Direction of the flat position”: “Storm: the rack moves flat. Repeat afterwards.” After the storm, tap REPEAT.

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5.2 The steps at a glance

NO.	STEP	1 AXIS	2 AXES	MOVES THE RACK?
1	Safety	✓	✓	no
2	System data	✓	✓	no
3	Board and power supply	✓	✓	no
4	Anemometer	✓	✓	no
5	Direction of the flat position	✓	✓	yes: briefly, then fully flat
6	Tilt travel	CALIBRATE TRAVEL TIME	fully steep and back	yes: whole way
7	Rotation east – west	–	✓	yes: whole way
(+1)	Reset of the sync controller	only if a reset output is selected under “System data”		yes: fully flat
7	Move to angle	✓	–	yes: to 30°
8	Sun sensor	south, north	east, west, south, north	no
9	Wi-Fi and time optional	✓	✓	no
10	Location optional	with orientation	without orientation	no
11	Lamps, sound, motion sensor optional	✓	✓	no
12	Test run of the automatic	Tilt	tilt, then rotation	yes: a little
13	Done	✓	✓	no

Both operating modes have 13 steps: with 2 axes “Rotation east – west” is added, with 1 axis “Move to angle”. If a reset of the sync controller is set (for example on the Samba Solar Rack with BST-KZQ-15A-2), there are 14, and from this step on the numbers shift by one. The header always shows “STEP n OF m”.

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5.3 Step by step

1 Safety both



Fig. 19 – Step 1: the safety notes.

WHAT YOU SEE

“The rack will move soon. Please check first” with four items and the ALL CHECKED button.

WHAT YOU DO

- Nobody in the swivel range, no animals either.
- No ladder, no tools, no cable in the way.
- The wind is light.
- The power supply at the rack is switched on.

Then ALL CHECKED.

WHAT MUST HAPPEN

Step 2 appears.

IF NOT

First put everything in order. This step cannot be skipped.

2 System data both

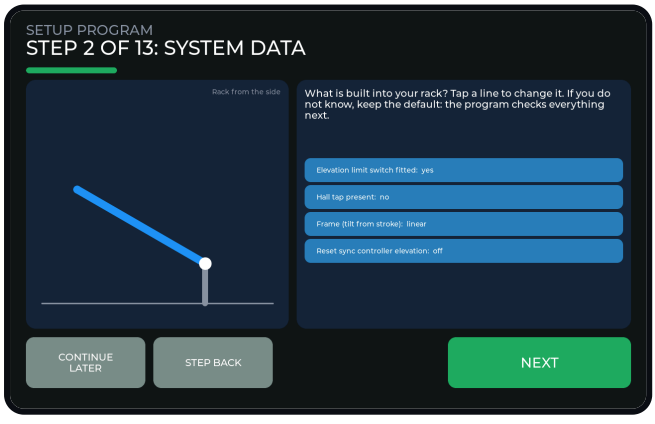


Fig. 20 – 1 axis four lines: elevation limit switch, Hall tap, frame, reset elevation.

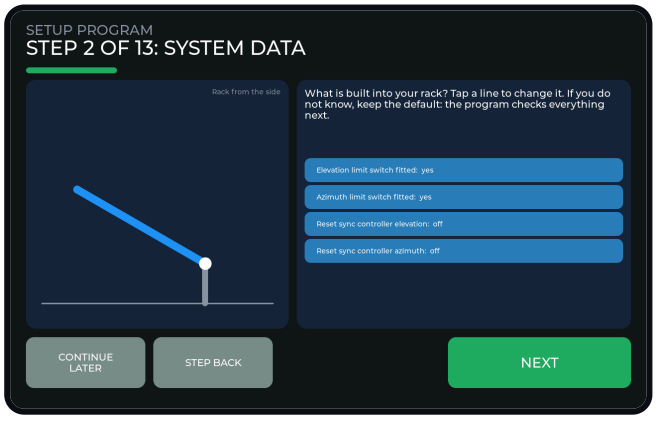


Fig. 21 – 2 axes limit switches of both axes, reset of both axes.

WHAT YOU SEE

Blue lines with what is fitted on your rack, each with its current value.

WHAT YOU DO

Tap each line until the right value is shown (each tap switches on). What the values mean and what

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is right for your rack: limit switches, Hall tap, frame, reset. Then NEXT.

WHAT MUST HAPPEN
The lines show your values. Step 3 appears.

IF NOT
If you do not know, keep the default. The following steps check it.

Samba Solar Rack 2.0-E Plus

Elevation limit switch: **no** · Hall tap: **yes**, if the HallTapBoard is fitted · Frame: **G2** · Reset sync controller elevation: **J23** (board from draft 0.12, sync controller BST-KZQ-15A-2).

Other rack

Limit switches: **yes** only if your own limit switches at the end of the travel path are connected to the board. Frame: **linear**. Reset: **off**, unless your sync controller has a reset input and it is connected.

3 Board and power supply both

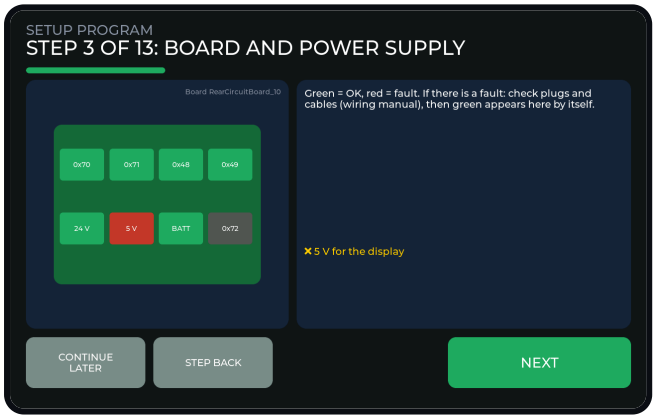


Fig. 22 – A fault: “5 V” is red, on the right it says “5 V for the display”.

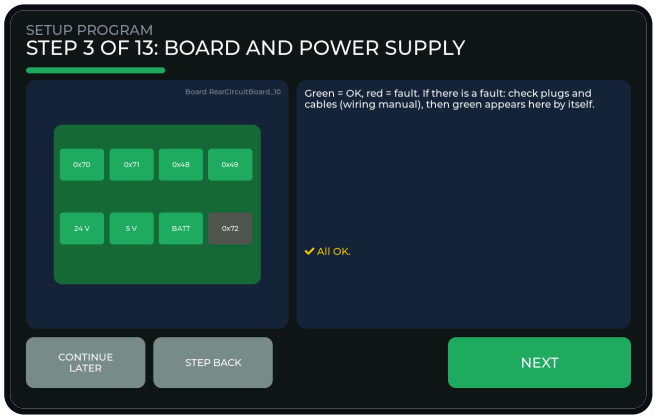


Fig. 23 – All OK: all fields green (“0x72” grey = not used).

WHAT YOU SEE
First an explanation with CHECK, then a picture of the board with eight fields.

WHAT YOU DO
Tap CHECK, wait a moment.

WHAT MUST HAPPEN
All fields green or grey, on the right “All OK”. Then NEXT.

IF NOT
Red field: check plugs and cables (wiring manual, German only). The field turns green by itself as soon as the fault is gone. Only then does NEXT work.

FIELD	WHAT IS CHECKED	RED MEANS
0x70	Chip of the switching outputs and limit switches	Board not seated properly or faulty; without it nothing moves
0x71	Chip of the indicator lights	Lights and supply monitoring faulty

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FIELD	WHAT IS CHECKED	RED MEANS
0x48	Measurement of the sun sensor	Chip faulty or sun sensor supply overloaded (short circuit in the cable?)
0x49	Measurement of the supply	second measuring chip not responding
24 V	24 V from the power supply	Controller power supply off or fuse
5 V	5 V for the display	Voltage converter of the board not delivering
BATT	Battery charge lock and sensor supply	Charge lock implausible (see manual for assembly and operation, Chapter 7)
0x72	Chip of the reset contacts (terminal J23)	grey if no reset at J23 is selected; red if it is selected but not responding (board before draft 0.12?)

4 Anemometer both

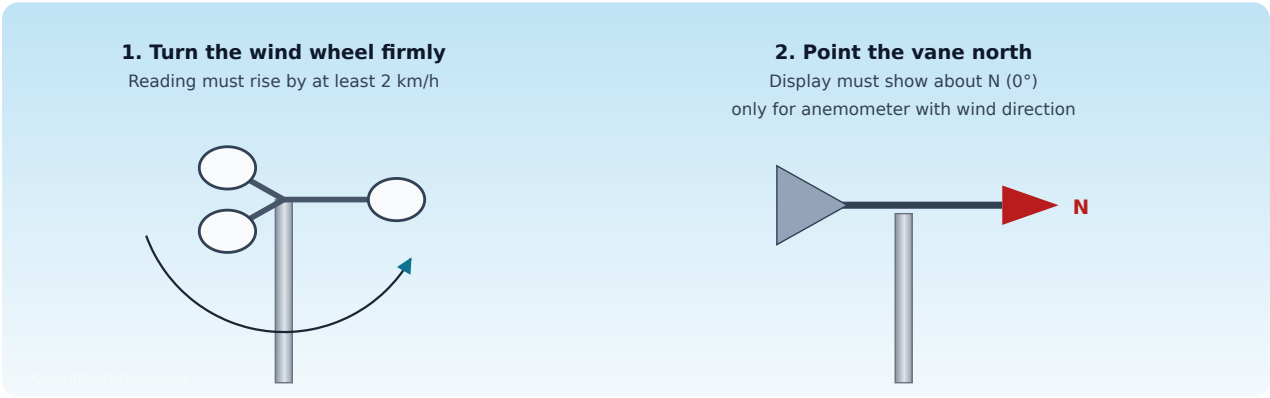


Fig. 24 – Two checks on the anemometer: turn the wind wheel; with an anemometer with direction, also point the vane north.

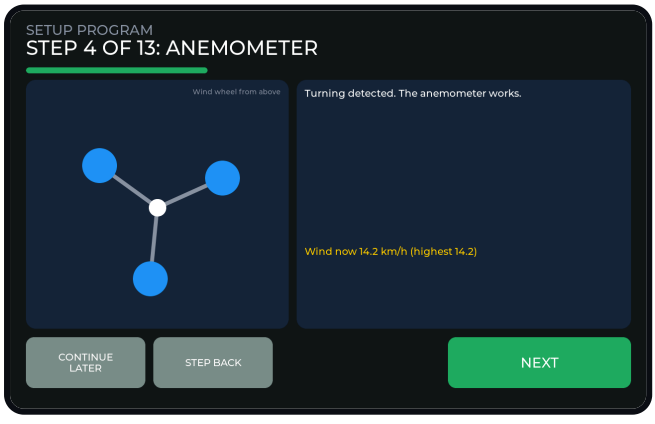


Fig. 25 – "Turning detected. The anemometer works."

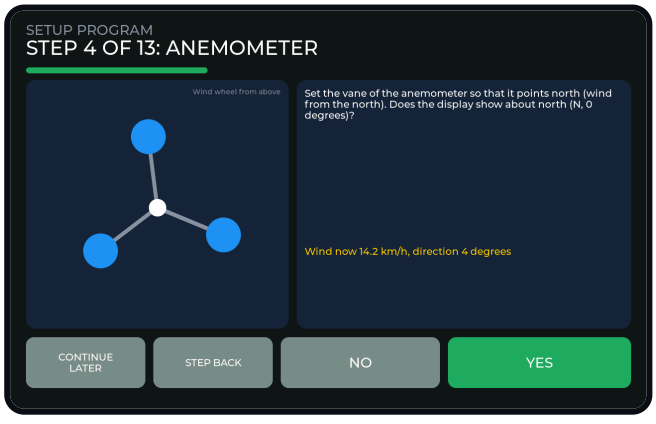


Fig. 26 – Question about the wind direction (only anemometer with direction).

WHAT YOU SEE

A wind wheel from above; at the bottom right in yellow "Wind now ... km/h (highest ...)".

WHAT YOU DO

1. Tap START.
2. Turn the wind wheel firmly by hand.
3. NEXT.

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4. Only with wind direction: point the vane north and see whether the display shows about N (0°); YES or NO.

WHAT MUST HAPPEN

When turning, the reading rises by at least 2 km/h, and the text changes to “Turning detected”. The direction shows about north.

IF NOT

“The wind wheel has not been turned yet.”: turn it more firmly. No number: the anemometer is not responding; check the wiring and the anemometer data. Wrong direction (NO): check the mounting and north mark of the anemometer, then REPEAT.

5 Direction of the flat position both

The most important step: it decides where the rack moves in a storm (Chapter 3).

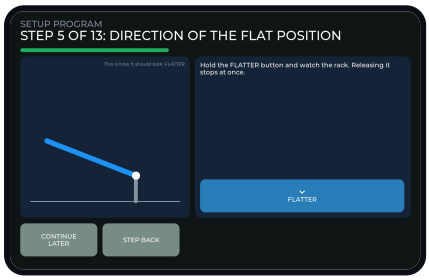


Fig. 27 – Hold the FLATTER button (5 s at most).

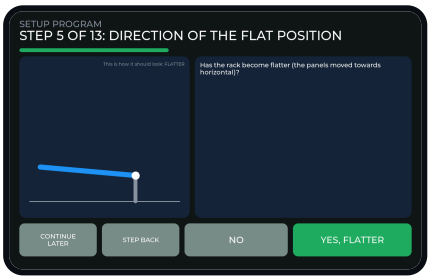


Fig. 28 – “Has the rack become flatter?”

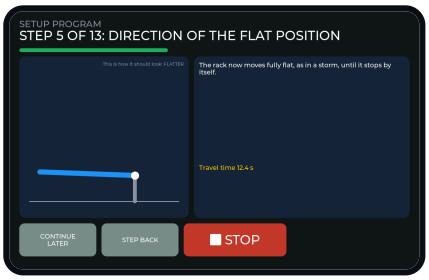


Fig. 29 – Then the rack moves fully flat, as in a storm. At the bottom the running travel time.

WHAT YOU SEE

On the left a sketch “This is how it should look: FLATTER”, on the right the instructions and a large FLATTER button.

WHAT YOU DO

1. NEXT.
2. Hold FLATTER and watch the rack. Releasing it stops at once; after 5 s it stops by itself.
3. YES, FLATTER if the panels have moved towards horizontal, otherwise NO.
4. Watch until the rack stops by itself.

WHAT MUST HAPPEN

The rack becomes flatter and then moves to the end position. “The flat position is checked. From now on the storm protection keeps the rack safe.”

IF NOT

With NO the device switches the direction itself (value “Flat position is in UP direction”): “Good that we checked”, then REPEAT. If nothing moved at all: check the power supply at the rack, the wiring of the contacts, the sync controller (momentary mode?). If the rack is already fully flat, the device asks straight away whether it is flat.

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⚠️ **Without limit switches the flat move takes the full maximum run time**

If “Elevation limit switch fitted” is set to no, the controller waits for the whole maximum run time, even if the rack has already stopped (run-time end position). On the Samba Solar Rack that is 90 s. With a Hall tap the move ends about 1 s after the pulses stop.

6 Tilt travel

1 axis CALIBRATE TRAVEL TIME

The device measures how long the rack takes from fully flat to fully steep and back. From this it later knows every angle. Tap **CALIBRATE TRAVEL TIME**; a separate screen guides you in four steps (pictures below). Then you come back and tap NEXT.

2 axes fully steep and back

START: the rack moves fully steep to the end stop, then fully flat to the end stop. The device measures both times. Then it asks whether the rack moved fully steep and fully flat again without catching or bumping.

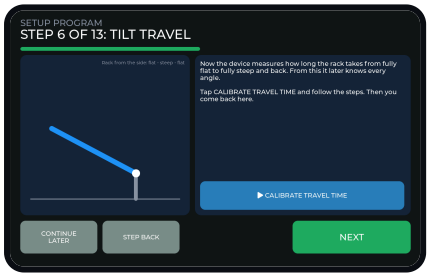


Fig. 30 – 1 axis step 6 with the blue CALIBRATE TRAVEL TIME button.



Fig. 31 – 2 axes “Moving STEEPER to the end stop …” with running time.



Fig. 32 – 2 axes the question about the move.

1 axis The four steps of CALIBRATE TRAVEL TIME

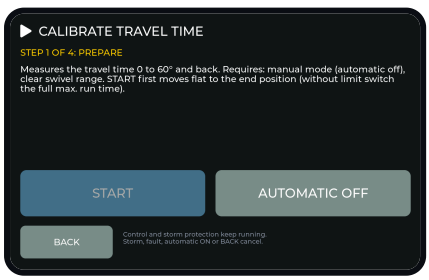


Fig. 33 – 1 of 4, prepare: START only works with the automatic switched off (AUTOMATIC OFF button).

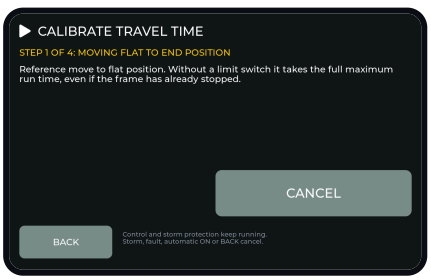


Fig. 34 – Reference move to the flat position. Without limit switches it takes the full maximum run time.

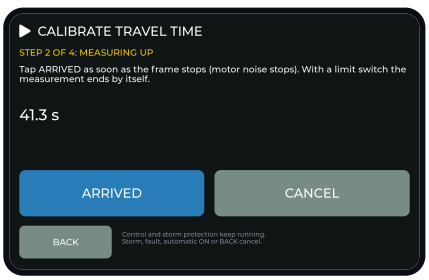


Fig. 35 – 2 of 4, measuring UP (steeper): tap ARRIVED as soon as the frame stops (motor noise stops). With limit switches or a Hall tap the measurement ends by itself.

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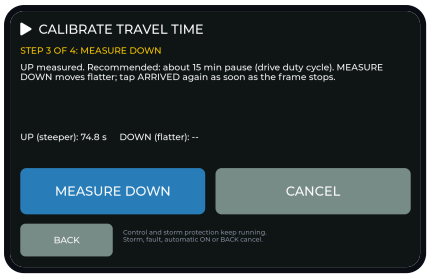


Fig. 36 – 3 of 4: UP measured. A pause of about 15 min is recommended (duty cycle). MEASURE DOWN moves flatter; tap ARRIVED again.

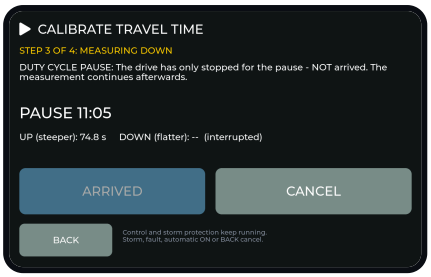


Fig. 37 – If the drive has only stopped for the duty cycle pause, it has NOT arrived. The measurement continues afterwards.

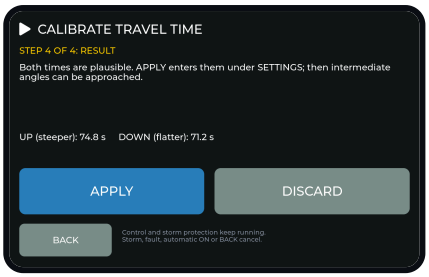


Fig. 38 – 4 of 4, result: APPLY enters both times under SETTINGS.

WHAT YOU SEE

1 axis the CALIBRATE TRAVEL TIME button.
2 axes the running travel time, then the question.

WHAT YOU DO

1 axis go through CALIBRATE TRAVEL TIME, APPLY, back, NEXT. 2 axes START, watch, YES or NO.

WHAT MUST HAPPEN

The rack reaches both end positions without catching or bumping. 1 axis "Both times are plausible": each at least 10 s, shorter than 0.9 × maximum run time, UP and DOWN at most in a ratio of 1 : 2.

IF NOT

"The travel time has not been calibrated yet.": calibrate first. Cancelled with "MAX. RUN TIME TOO SHORT OR ARRIVED NOT TAPPED": increase the maximum run time. NO with 2 axes: check mechanics and drive (assembly manual of the rack), then REPEAT.

ℹ️ **With Hall tap** 1 axis

If "Hall tap present" is set to yes, the line is called "CALIBRATE: TRAVEL TIME AND HALL STEPS". The measuring moves then end by themselves and count the Hall steps; plausible are at least 1000 steps each, UP and DOWN at most 3 % apart. If the rejection reads "HALL COUNTS BACKWARDS", set "Invert Hall count direction" to yes.

7 Rotation east – west 2 axes

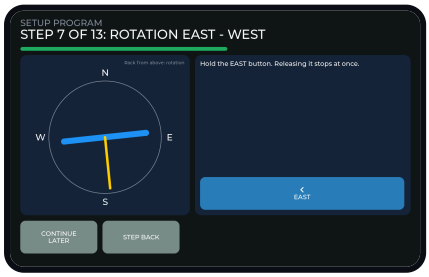


Fig. 39 – Hold the EAST button.

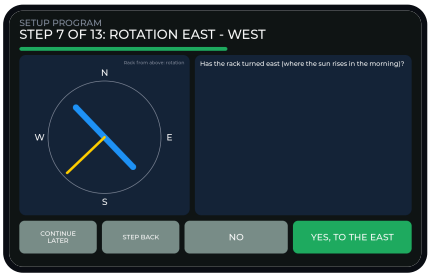


Fig. 40 – "Has the rack turned east?"

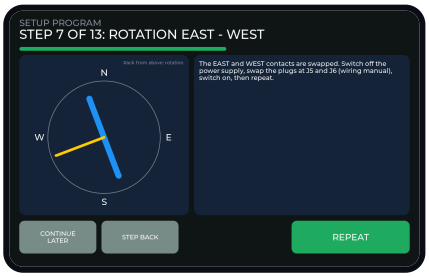


Fig. 41 – With NO: the EAST and WEST contacts are swapped.

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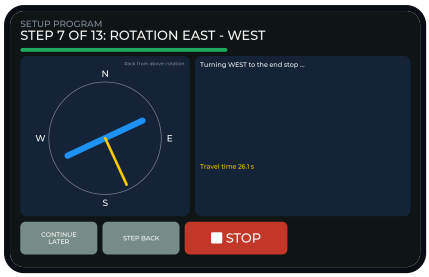


Fig. 42 – Then travel path east → west → east to the end stops.

WHAT YOU SEE

On the left a top view with a compass (yellow: direction the panels face), on the right the instructions and the EAST button.

WHAT YOU DO

1. NEXT, then hold EAST (5 s at most).
2. YES, TO THE EAST if the rack has turned towards the morning sun.
3. Watch: the rack turns fully east, fully west and back east.

WHAT MUST HAPPEN

All three moves reach their end stop. “Rotation checked. The rack is in the morning position.”

IF NOT

NO: this cannot be changed by a setting. Switch off the power supply, swap the plugs at J5 and J6 (wiring manual, German only), switch on, REPEAT. No movement: check the power supply and wiring of the azimuth drive.

+ Reset of the sync controller only if set

This step only exists if a reset output is selected in step 2. It comes after the travel path (with 2 axes after the rotation). There is no picture of its own for it yet; the band in the dashboard shows the progress, as with a reset via SETTINGS.

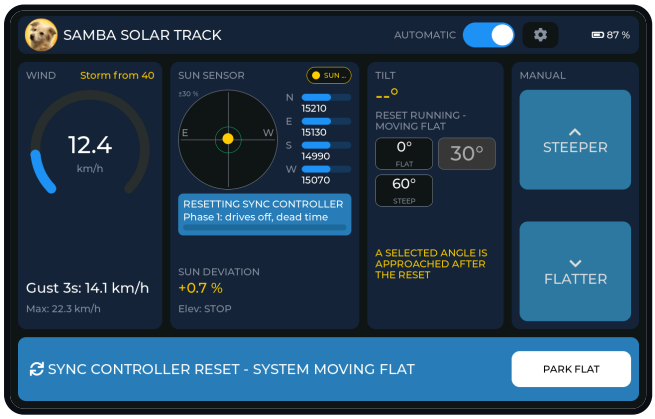


Fig. 43 – How the device shows a running reset (picture from the sequence via SETTINGS).

WHAT YOU SEE

“The sync controller of the drives has a reset input. In a moment the device triggers it.” and the TRIGGER RESET button.

WHAT YOU DO

TRIGGER RESET, watch. With 2 axes and two resets, both run one after the other.

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WHAT MUST HAPPEN

The sync controller fully retracts the drives (on the rack: flat). “Reset checked.”

IF NOT

“Reset not possible (check connection or chip).”: it did not start within 20 s; check terminal J23 or J5 and the value reset. “Reset aborted”: the cause is in the status bar; then REPEAT.

Assumption

The duration of the reset pulse (5 s) and the behaviour of the sync controller afterwards have not yet been measured on the rack.

7 Move to angle 1 axis

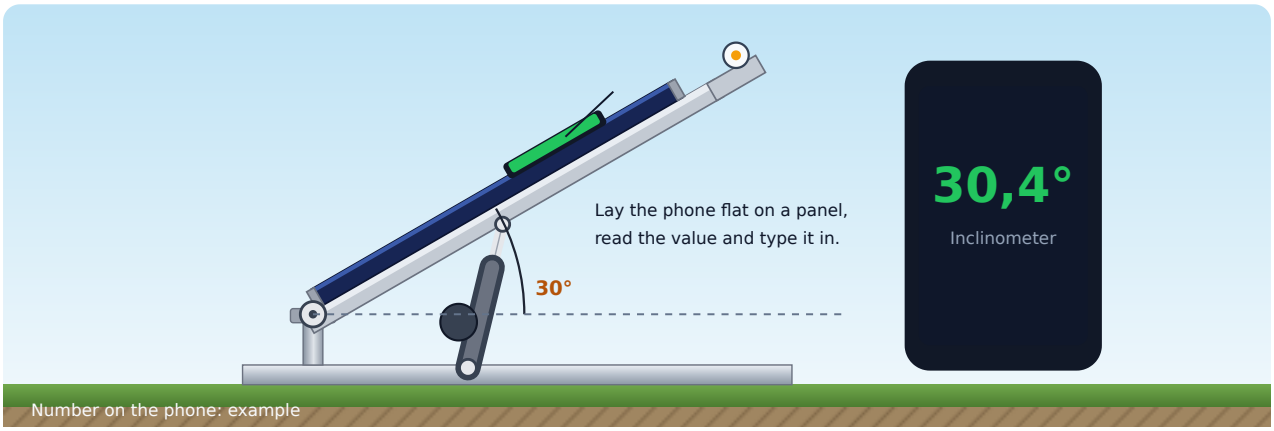


Fig. 44 – The rack is at 30°. A phone with an inclinometer app lies flat on a panel.

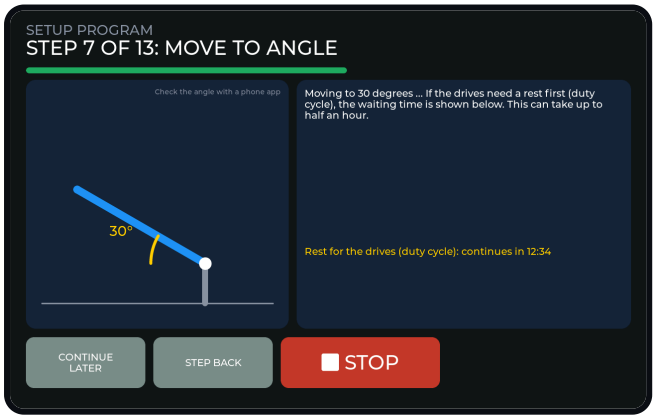


Fig. 45 – “Moving to 30 degrees ...”

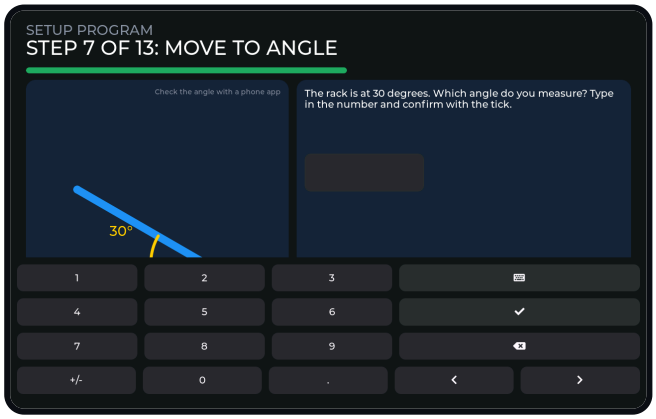


Fig. 46 – Type in the measured angle and confirm with the tick.

WHAT YOU SEE

“Now the rack moves to 30 degrees.” and the MOVE TO 30 DEGREES button.

WHAT YOU DO

Tap the button, wait. Then lay the phone flat on a panel, read the angle, type it in (0 to 90), tick.

WHAT MUST HAPPEN

IF NOT

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The rack stops at about 30°. “Angle recorded. It is in the test report.”

“Angle cannot be reached (travel time or Hall pickup).”: repeat step 6. If your value differs a lot: check the frame. There is no limit yet beyond which the step fails.

8 Sun sensor

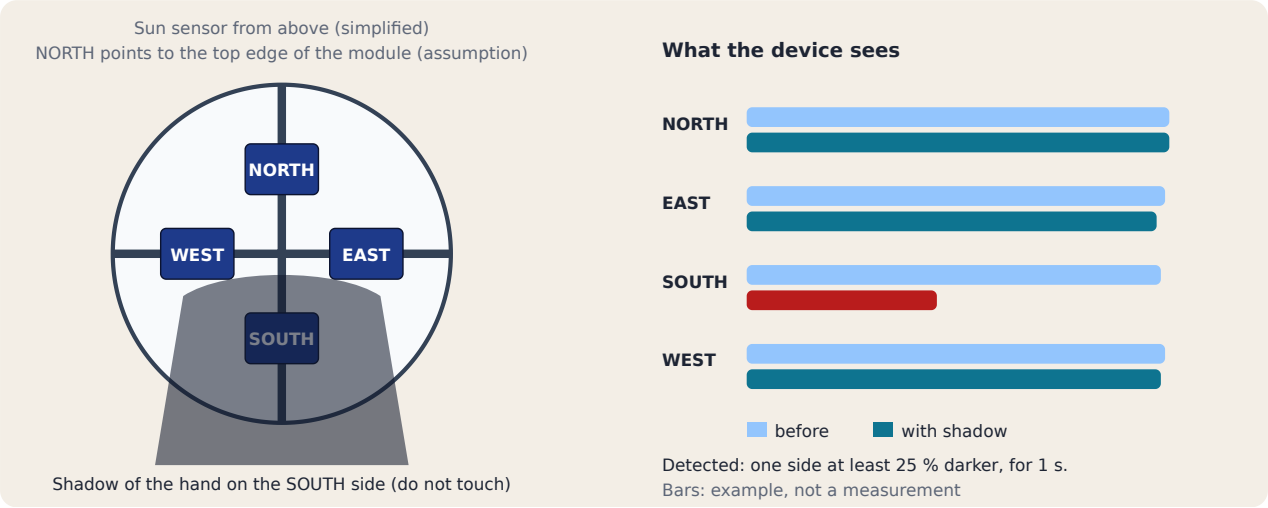


Fig. 47 – Shade one side of the sun sensor: hold your hand over it without touching it. The device detects which input gets darker.

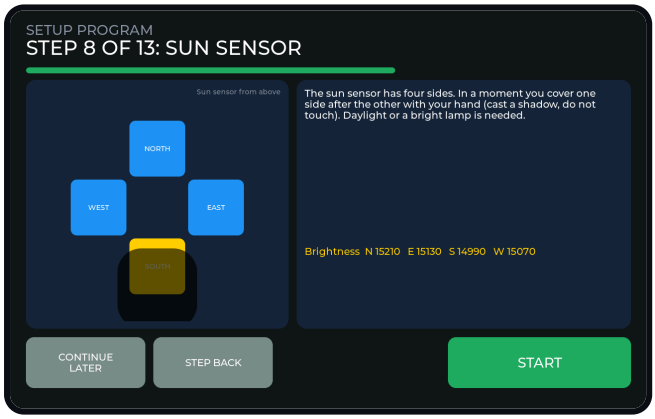


Fig. 48 – Start: at the bottom the brightness of all four sides.

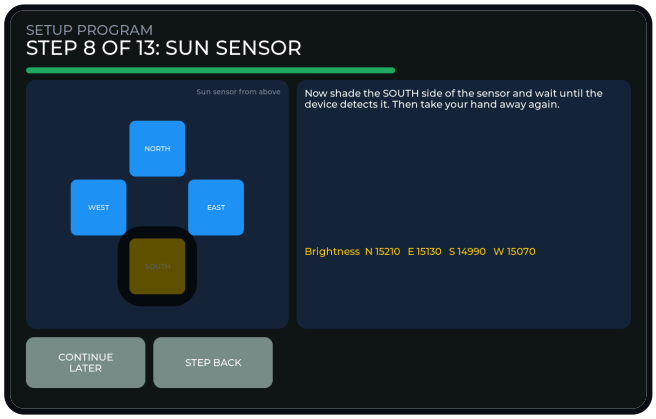


Fig. 49 – “Now shade the SOUTH side of the sensor” (picture with 2 axes).

WHAT YOU SEE

The sensor from above with four fields. The side you are to shade is marked.

WHAT YOU DO

START. Then shade the named side, wait until the device detects it, take your hand away. **1 axis** SOUTH, then NORTH. **2 axes** EAST, WEST, SOUTH, NORTH.

WHAT MUST HAPPEN

Each side becomes at least 25 % darker and is detected for 1 s. “All sides of the sensor are assigned correctly.”

IF NOT

If the device detects a different side, it corrects the assignment of the inputs itself and records this in the test report. “Too dark”: repeat in daylight or with a bright lamp. Nothing is

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detected: make the shadow larger, closer to the sensor.

9 Wi-Fi and time optional

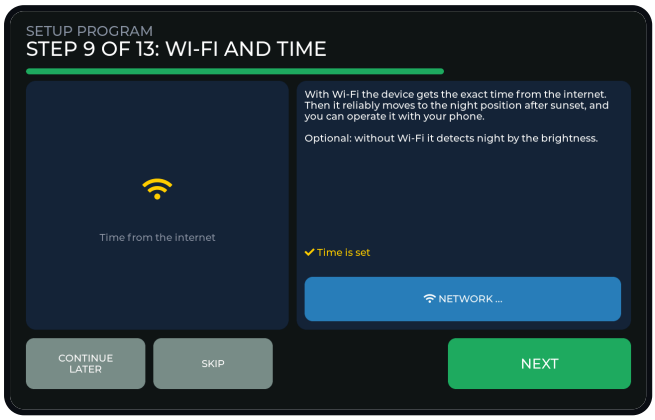


Fig. 50 – “Time is set”: Wi-Fi is already connected.

WHAT YOU SEE
An explanation, the NETWORK ... button and NEXT.

WHAT YOU DO
NETWORK ... opens the screen from Chapter 4.4. After the restart the setup program opens again at this point. Or SKIP.

WHAT MUST HAPPEN
“Time is set”, then NEXT.

IF NOT
“No time yet: set up Wi-Fi or skip.” Without Wi-Fi the device detects night by the brightness.

10 Location optional

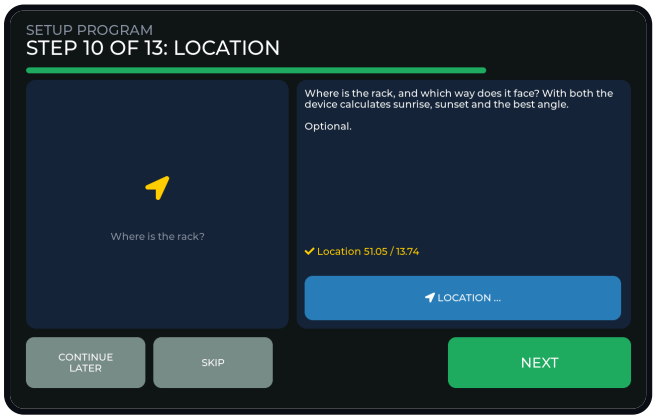


Fig. 51 – Location entered: at the bottom “Location 51.05 / 13.74”.

WHAT YOU SEE
1 axis “Where is the rack, and which way does it face?” 2 axes “Where is the rack?” and the LOCATION ... button.

WHAT YOU DO
LOCATION ... opens the map (Chapter 6.2); with 1 axis afterwards also the orientation. Or SKIP.

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WHAT MUST HAPPEN

At the bottom “Location …” with latitude and longitude, then NEXT.

IF NOT

“No location yet: enter it or skip.”

11 Lamps, sound, motion sensor optional

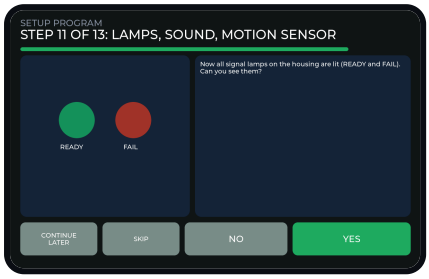


Fig. 52 – All indicator lights are lit: “Can you see them?”

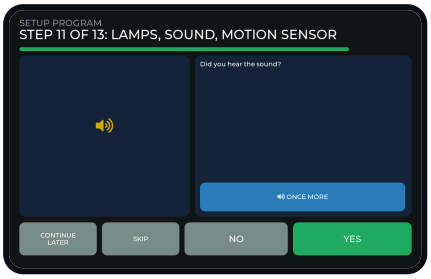


Fig. 53 – “Did you hear the sound?” ONCE MORE plays it again.

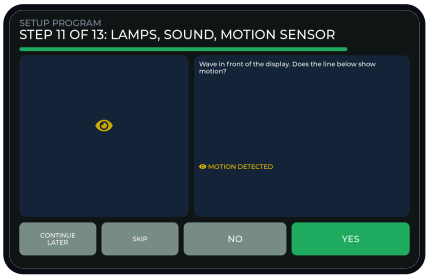


Fig. 54 – “Wave in front of the display.” At the bottom “MOTION DETECTED”.

WHAT YOU SEE

Three questions one after the other: lamps, sound, motion sensor.

WHAT YOU DO

START, then YES or NO each time. Or SKIP.

WHAT MUST HAPPEN

BEREIT (ready) and FAIL light up on the housing and on the board; the sound can be heard; when waving, “MOTION DETECTED” appears.

IF NOT

NO is noted in the test report, the step continues anyway. Lamps: check the connection of the housing lights (wiring manual, German only). Sound: volume. Motion: enough light in front of the display?

12 Test run of the automatic

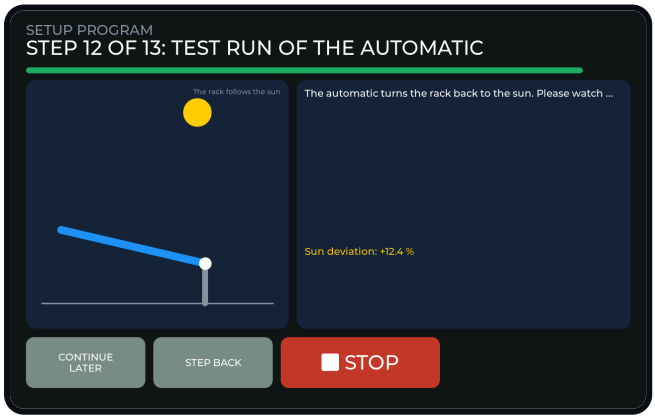


Fig. 55 – “The automatic turns the rack back to the sun.” At the bottom the sun deviation.

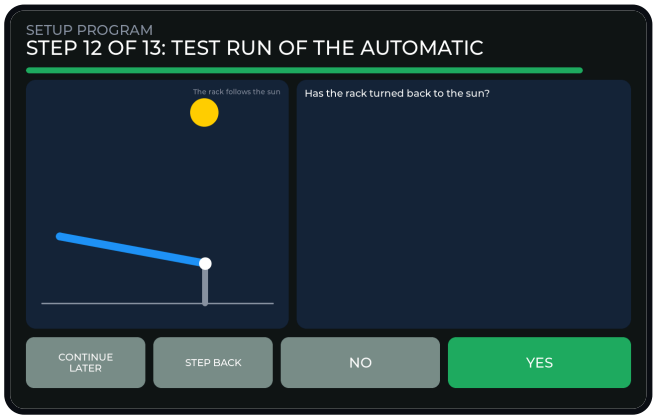


Fig. 56 – “Has the rack turned back to the sun?”

WHAT YOU SEE

An explanation, then “Moving the rack off a little …”, then the running automatic with the deviation in percent.

WHAT YOU DO

START, watch, YES or NO. **2 axes** first the tilt, then the rotation.

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WHAT MUST HAPPEN
The device moves the axis off for about 4 s, then the automatic moves back to the sun and stops. “The test run is passed.”

IF NOT
If the automatic moves away from the sun, the device switches the direction itself (“The direction has been switched. Please once more.”): for the tilt “North brighter means UP”, for the rotation it swaps the east and west inputs. “The automatic mode did not align. Is there enough sun?”: after 3 min without success; repeat when the sun is out.

13 Done

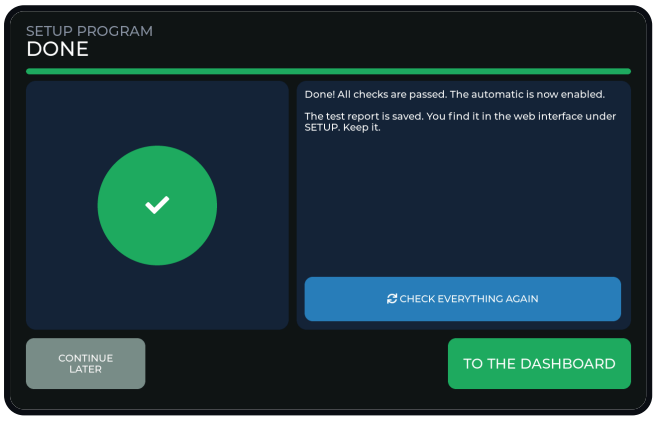


Fig. 57 – “Done! All checks are passed. The automatic is now enabled.” Below it the CHECK EVERYTHING AGAIN button; it asks for confirmation first (5.4).

WHAT YOU SEE
A green tick, TO THE DASHBOARD and the CHECK EVERYTHING AGAIN button.

WHAT YOU DO
TO THE DASHBOARD. Switch on the automatic switch at the top if it is off.

WHAT MUST HAPPEN
The purple band is gone. The status bar shows the automatic (for example “AUTO: ALIGNED”).

IF NOT
A mandatory step is still missing: “This step is still missing.” Go back to that step.

5.4 After completion

- **Test report:** the device saves the last three reports. In the web interface under SETUP → SETUP PROGRAM, VIEW shows the report as a designed sheet (light for the printer, dark for a file) with the button SAVE AS PDF; TEXT downloads it as a text file (since firmware 0.7.6, older reports as text only). It contains date, firmware, operating mode, anemometer, every step with its result and the measured values (travel times, Hall steps, angle read off, assignment of the sun sensor, answers on lamps, sound and motion sensor). Keep it: it helps with later checks and queries.
- **Individual values** can still be changed under SETTINGS (Chapter 7); CALIBRATE TRAVEL TIME can be repeated there at any time **1 axis**.
- **Check everything again:** after replacing a part (drive, sensor, anemometer, board) or after changing the operating mode: SETTINGS → COMMISSIONING → SETUP PROGRAM (CHECK

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EVERYTHING) opens the “Done” screen; there tap CHECK EVERYTHING AGAIN and START OVER in the confirmation prompt. The program starts at step 1. Until all steps are passed again, the automatic is locked; storm protection and parking keep working. The previous test report remains saved. A setup program completed only in the simulation starts again by itself on the real rack (4.10).

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6 Location, orientation, sun and night

If the device knows the time (via Wi-Fi) and the location, it calculates the course of the sun: sunrise, sunset, height and direction of the sun. With this it moves to the night position on time. During the day the sun sensor still does the tracking.

> **Since firmware 0.7.0, tested on the PC only**
> The sun position calculation has been tested against reference values from an astronomical library. It has not yet been tried on the device with real time and a real rack.

6.1 The SUN page

On the dashboard, at the top of the SUN SENSOR card, there is a small yellow field with sunrise and sunset, for example “↑ 04:50 ↓ 21:23”. Tapping it opens the SUN page. When it opens, the sun runs in a short animation from sunrise to its current position.

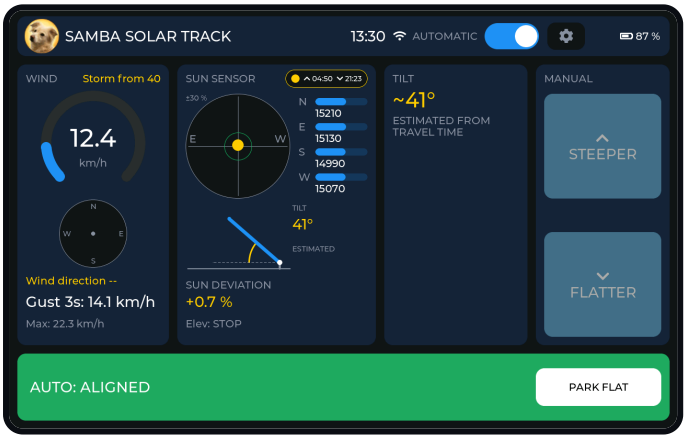


Fig. 58 – 1 axis dashboard: the yellow field at the top of the SUN SENSOR card opens the SUN page.



Fig. 59 – 2 axes the same field on the dashboard with two axes.



Fig. 60 – 1 axis SUN: arc of the day, times, best tilt angle and orientation of the rack.



Fig. 61 – 2 axes SUN: instead of the orientation, the night position “FLAT, THEN EAST” and the NIGHT TO THE EAST button.

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Fig. 62 – On opening: the sun runs along the arc (picture in the middle of the animation).

ITEM	MEANING
Orange line	Path of the sun across the sky today, from north-east via south to north-west (in summer). Yellow dot: the sun now.
NOON	Time and height of the highest sun position.
SUN ELEVATION, DIRECTION	How high (in degrees above the horizon) and in which compass direction the sun is now.
SUNRISE / SUNSET	Today, for your location.
NIGHT FROM, DAY FROM	When the device moves to the night position and when the day begins (with the offsets from 6.4).
BEST TILT ANGLE **1 axis**	Which tilt catches the most sun now, taking the orientation of the rack into account.
RACK FACES **1 axis**	The set orientation (6.3).
AT NIGHT **2 axes**	Night position: “FLAT, THEN EAST” or only flat. The NIGHT TO THE EAST button switches between them.

6.2 Entering the location

You reach the location via the LOCATION … button on the SUN page or in the setup program (step 10). For the sun position an accuracy of about 0.1° (roughly 10 km) is enough: the nearest town will do.

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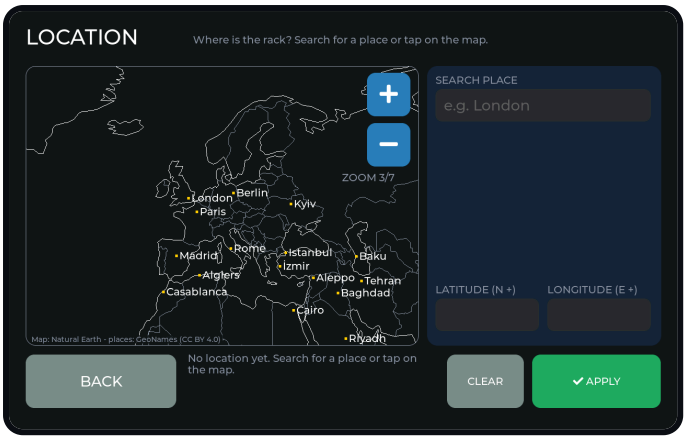


Fig. 63 – No location yet: map of Europe, empty fields.

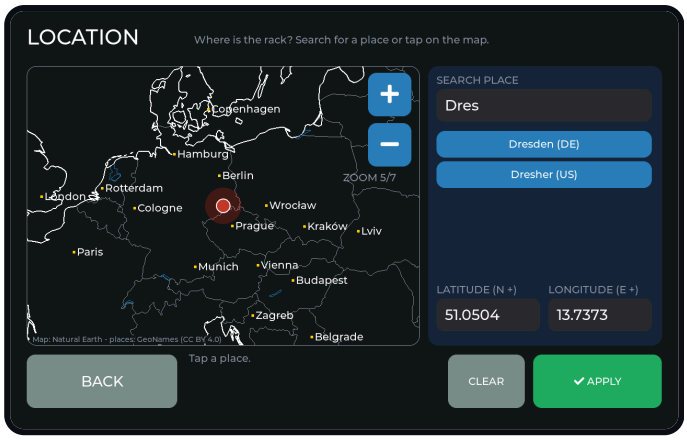


Fig. 64 – Place search: “Dres” typed, suggestions “Dresden (DE)” and “Dresher (US)”.

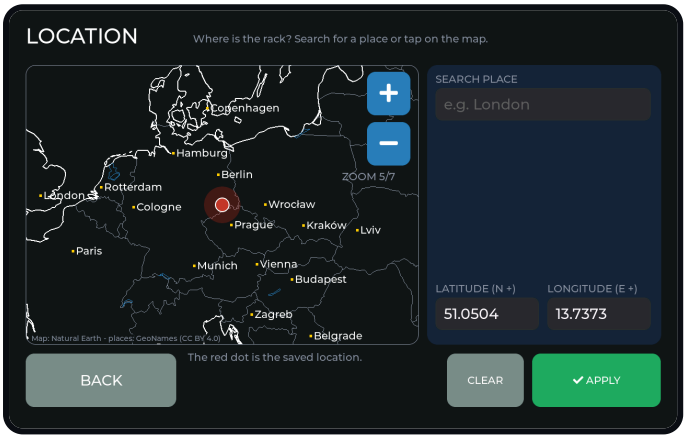


Fig. 65 – Saved: the red dot marks the location, below it latitude and longitude.

1

Search for a place or tap on the map

Type the first letters of your town into the SEARCH PLACE field and tap a suggestion. Or enlarge the map with + and −, move it with your finger and tap your town. *Result:* Latitude and longitude are in the fields.

2

Check or type in the numbers

If you know the coordinates (for example from a map app), type them in directly: LATITUDE (N +), LONGITUDE (E +), in decimal degrees with a dot, for example 51.0504 and 13.7373. South of the equator the latitude is negative, west of Greenwich the longitude. *Result:* The numbers are right.

3

APPLY

Result: The red dot sits on your town; the SUN page shows the times for your location.

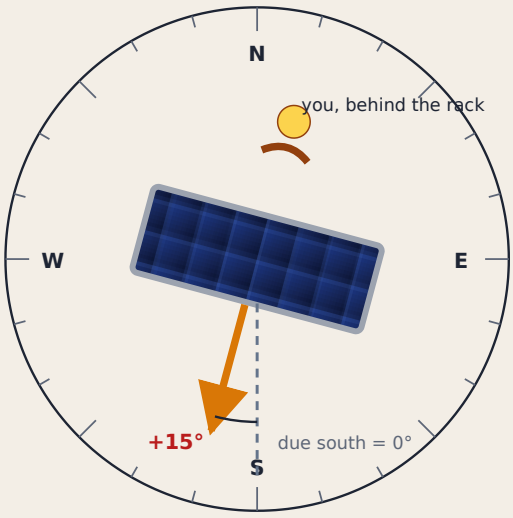
ℹ️ **Latitude 0 and longitude 0 means “not entered”**

By default both are 0. CLEAR resets the location to “not entered”. The map on the display is stored in the device and needs no internet (outlines from Natural Earth, places from GeoNames). The web interface also has a street map (Chapter 8).

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6.3 Orientation of the rack 1 axis

A rack with one axis cannot turn. If it does not face exactly south, the sun is “in front” of the rack at different times. So that the device calculates correctly, you tell it which way your rack faces. With two axes this value is not needed, because the azimuth follows the sun anyway.



How to read the value

Rack viewing direction	Value
south-east (SE)	−45°
south-south-east (SSE)	−22.5° → −22 or −23
due south (S)	0°
south-south-west (SSW)	+22.5° → +22 or +23
south-west (SW)	+45°

Turned west: **plus**. Turned east: **minus**.
Allowed: −90° to +90°, in whole degrees.
A few degrees of compass error do not matter.
Do not hold the compass right against the rack's metal.

Only needed with one axis

Fig. 66 – Measuring the orientation: stand behind the rack and look in the same direction as the panels. If the viewing direction deviates from south towards the west, the value is positive; towards the east, it is negative. Here: 15° to the west = +15.

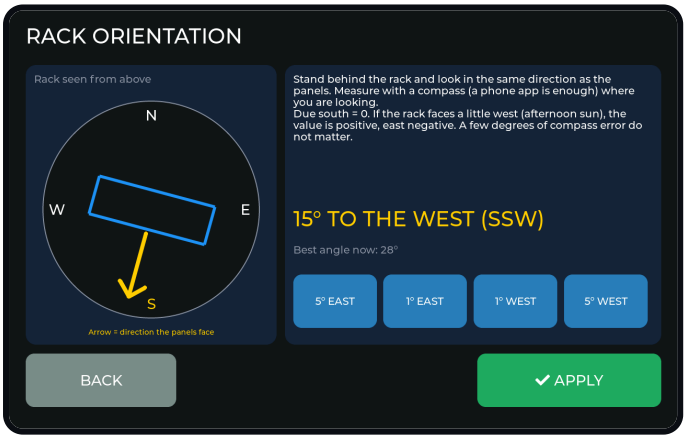


Fig. 67 – RACK ORIENTATION: the rack in the drawing turns along, below are the value and the best angle now.

1 Measure the direction

Measure with a compass (a phone app is enough) behind the rack, looking in the direction of the panels. Do not hold the compass right against the metal. *Result:* a compass direction, for example 195° (south-south-west).

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2

Convert to the value

Value = compass direction − 180. Example: 195° − 180 = +15. At 165° (south-south-east) this gives −15. *Result:* a number between −90 and +90.

3

Set it

SUN page → RACK ORIENTATION …, set with 5° EAST, 1° EAST, 1° WEST, 5° WEST, APPLY. *Result:* the SUN page shows “RACK FACES 15° TO THE WEST”.

Samba Solar Rack

The value depends only on how you set up the rack. There is no preset for the rack other than 0 (due south). A few degrees of compass error do not matter.

If the time or location is missing



Fig. 68 – Without time or location, the field only shows “SUN …”.

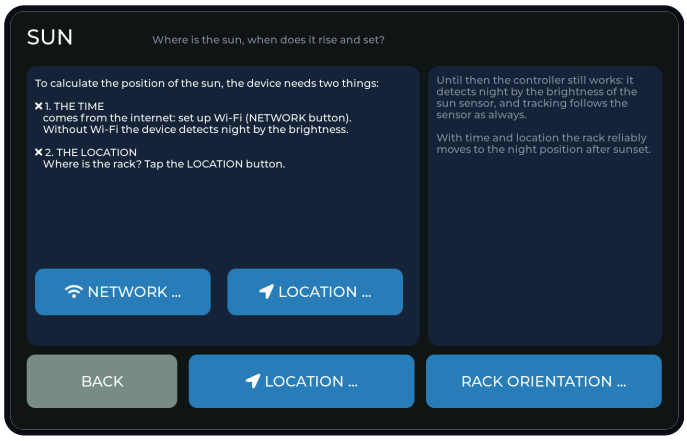


Fig. 69 – The SUN page says what is missing and offers NETWORK … and LOCATION ….

The controller still works fully: it then detects night by the brightness, and tracking follows the sun sensor as always.

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6.4 Night and night position

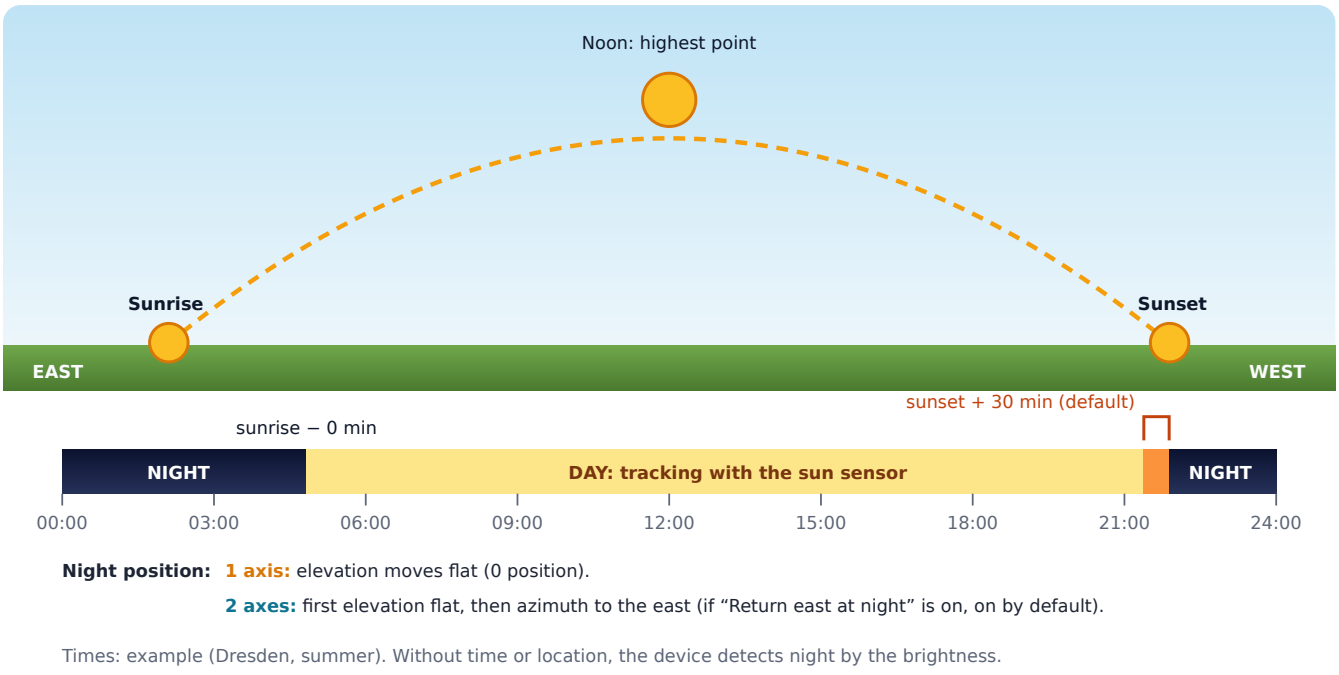


Fig. 70 – The day between sunrise and sunset. By default, night begins 30 min after sunset and ends at sunrise.

WHAT THE DEVICE KNOWS	NIGHT BEGINS	DAY BEGINS
Time **and** location	Sunset + “Night from sunset plus” (default 30 min), without waiting time	Sunrise − “Day from sunrise minus” (default 0 min)
Time or location missing	Brightness (sum of the four sides) below “Night below brightness” (default 4000) for as long as “Night or day only after” (default 5 min)	above “Day above brightness” (default 6000) for the same time

1 axis Night position

The elevation moves flat (0 position) and stays there until morning.

2 axes Night position

First the elevation moves flat, then the azimuth turns to the east (morning position) if “Return east at night” is on (on by default). Otherwise the azimuth stays where it is. The two axes never move at the same time.

The night move is a normal move with all protective functions; for example, it waits for a duty cycle pause to end. Storm, parking and manual mode take priority.

6.5 When cloudy, move to the calculated angle 1 axis off by default

Under thick cloud, the sun sensor cannot see a direction. If this switch is on, the elevation then moves to the calculated best angle (as with tilt on command), and the status bar shows “CLOUDY:

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…”. Requirements: time, location and a calibrated travel time. “Cloudy” means: the brightness sum stays below “Cloudy below brightness” (default 20000) for 5 min; it counts as sunny again above 1.25 times that value. The device only moves to a new target if it differs from the old one by at least 3°.

> 🧩 **Assumption**
> The threshold of 20000 is an estimate. On an overcast day, watch the brightness in the SUN SENSOR card (sum of the four values) and set the threshold slightly above it (Chapter 7).

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7 All adjustable values

The reference part: every value under SETTINGS with its meaning, default, limits and when it applies. In addition, a recommendation for the Samba Solar Rack and instructions on how to find the right value for a different installation. All values at a glance: Appendix A.

> ⚠️ **Check the details for the Samba Solar Rack**
> The recommendations for the rack come from the project description (README) and refer to the Rack 2.0-E Plus with the BST-KZQ-15A-2 sync controller and without limit switches on the tracker. They must be checked against the current version of the assembly manual on the website. Where a detail is missing, it says “measure on your own rack” or “default”; dimensions and times are not estimated here.

How to change a value

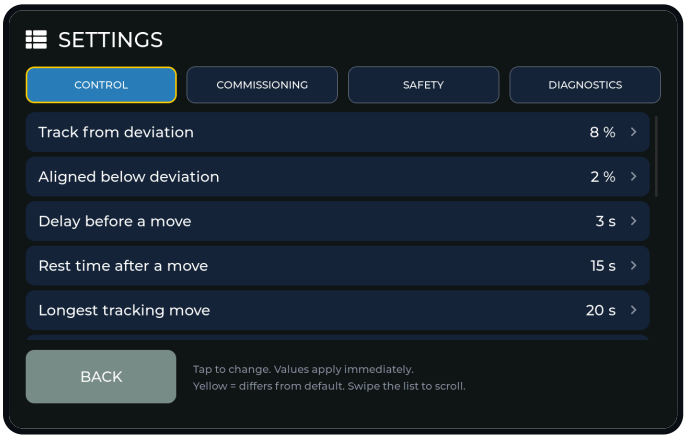


Fig. 71 – SETTINGS with four groups at the top. Each row shows the name and the current value.

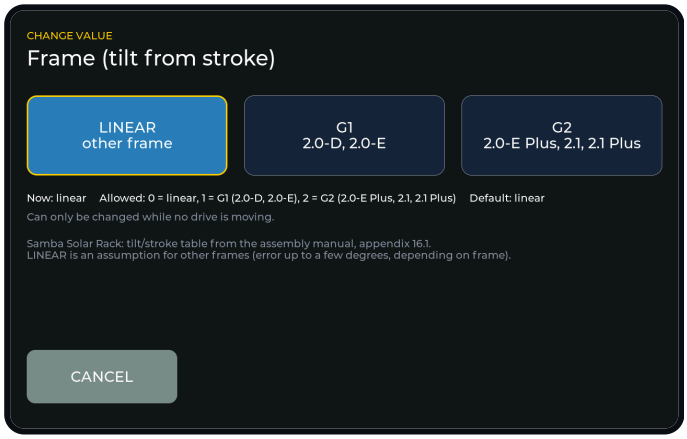


Fig. 72 – Tapping a row opens “CHANGE VALUE”, here for the frame: for a choice, tap one of the buttons (LINEAR, G1, G2); for numbers, type in the number. Below are the current value, the allowed values and the default (“Default”).

1 Gear icon → SETTINGS …

The button only exists once the setup has been saved. *Result:* the list of the CONTROL group appears.

2 Choose a group

CONTROL, COMMISSIONING, SAFETY or DIAGNOSTICS. Swipe the list to scroll. *Result:* the values of the group are shown; those in yellow differ from the default.

3 Tap a row, set the value, apply

Result: the row shows the new value. If it is outside the limits or does not fit another value, the device rejects it and says why.

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“APPLIES”	MEANING
immediately	The new value takes effect in the next control step.
only while drives are stopped	The change is rejected while a drive is moving. Wait a moment or wait for PARK FLAT to finish, then try again.
on the display immediately; remotely only after confirmation	If the value comes from the browser or ioBroker, the display shows a confirmation prompt with the old and new value. Only APPLY makes it valid. Proposals do not survive a restart.

Not every value appears on every installation: azimuth values only with 2 axes; travel times, frame, Hall tap, preset angles, orientation and cloudiness only with 1 axis; wind direction values only with an anemometer with direction. The tags show this.

SAFETY

Safety values protect the mechanics and drives. Their limits are fixed. If changed remotely, a new value only applies once you confirm it on the display (the display shows the old and new value).

Storm threshold (gust) both `stormKmh`

Default: **40 km/h** Allowed: **20 to 60 km/h**

Applies: **on the display immediately; remotely only after confirmation on the display**

From this wind speed on, the rack moves flat. What is measured is the gust: the average over 3 seconds.

SAMBA SOLAR RACK
Keep the default of 40 km/h as long as the rack's assembly manual does not give a different value.

OTHER RACK
Look in the data sheet of your frame for the permitted wind speed in operation (not parked) and stay well below it. If in doubt, set it lower: an unnecessary parking only costs a little yield.

Hold time after storm both `stormHoldMs`

Default: **10 min** Allowed: **5 to 60 min**

Applies: **on the display immediately; remotely only after confirmation on the display**

The rack stays flat for this long after the last gust above the storm threshold. Every new gust restarts the time.

SAMBA SOLAR RACK
Default 10 min.

OTHER RACK
In gusty wind, set it longer so that the rack does not constantly move back and forth (protects the drives).

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Dead time on direction change

both `motorDeadtimeMs`

Default: **500 ms** Allowed: **300 to 5000 ms**

Applies: **on the display immediately; remotely only after confirmation on the display**

Pause before a drive may change direction. Protects motor and gearbox.

SAMBA SOLAR RACK
Default 500 ms.

OTHER RACK
If the instructions of your drive or sync controller state a minimum pause when reversing polarity, use it; otherwise keep 500 ms.

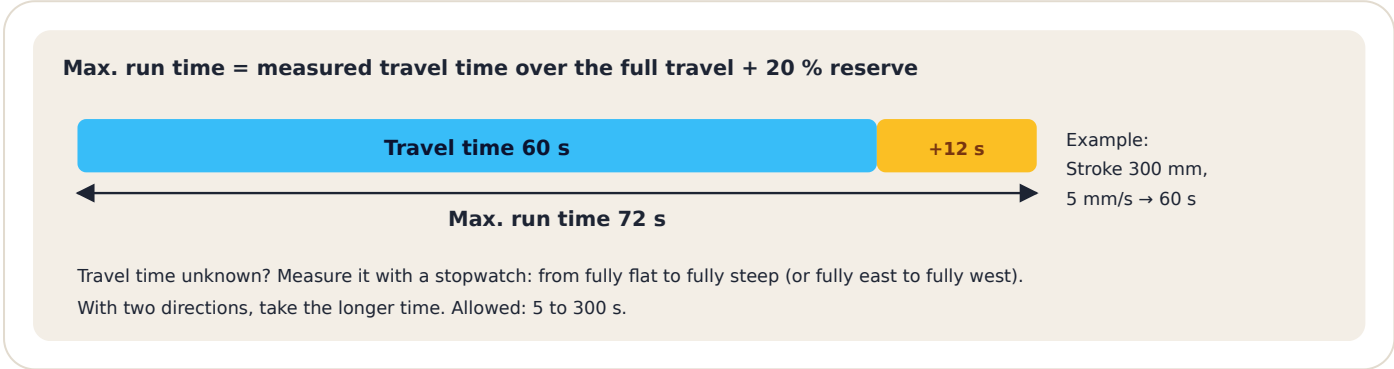


Fig. 73 – How to find the maximum run time. The example is taken from the Assembly and Operation Manual.

Max. run time elevation

both `motorMaxRunMsElevation`

Default: **60 s** Allowed: **5 to 300 s**

Applies: **on the display immediately; remotely only after confirmation on the display**

Longest permitted continuous move. With limit switches: if it is exceeded, the controller reports a drive fault. Without limit switches: when it has elapsed, the end position counts as reached (run-time end position).

SAMBA SOLAR RACK
90 s (Rack 2.0-E Plus according to the README).

OTHER RACK
Measure the travel time over the full travel (stopwatch, from fully flat to fully steep) and add 20 %. Or: stroke ÷ speed + 20 % (figure below).

Max. run time azimuth

2 axes `motorMaxRunMsAzimuth`

Default: **90 s** Allowed: **5 to 300 s**

Applies: **on the display immediately; remotely only after confirmation on the display**

As above, for the east–west rotation.

SAMBA SOLAR RACK
– (the rack has no azimuth).

OTHER RACK
Measure the travel time from fully east to fully west and add 20 %.

COMMISSIONING

What is fitted to your rack and how. The setup program sets most of these values itself. Values marked “only while drives are stopped” are rejected by the device while a drive is moving.

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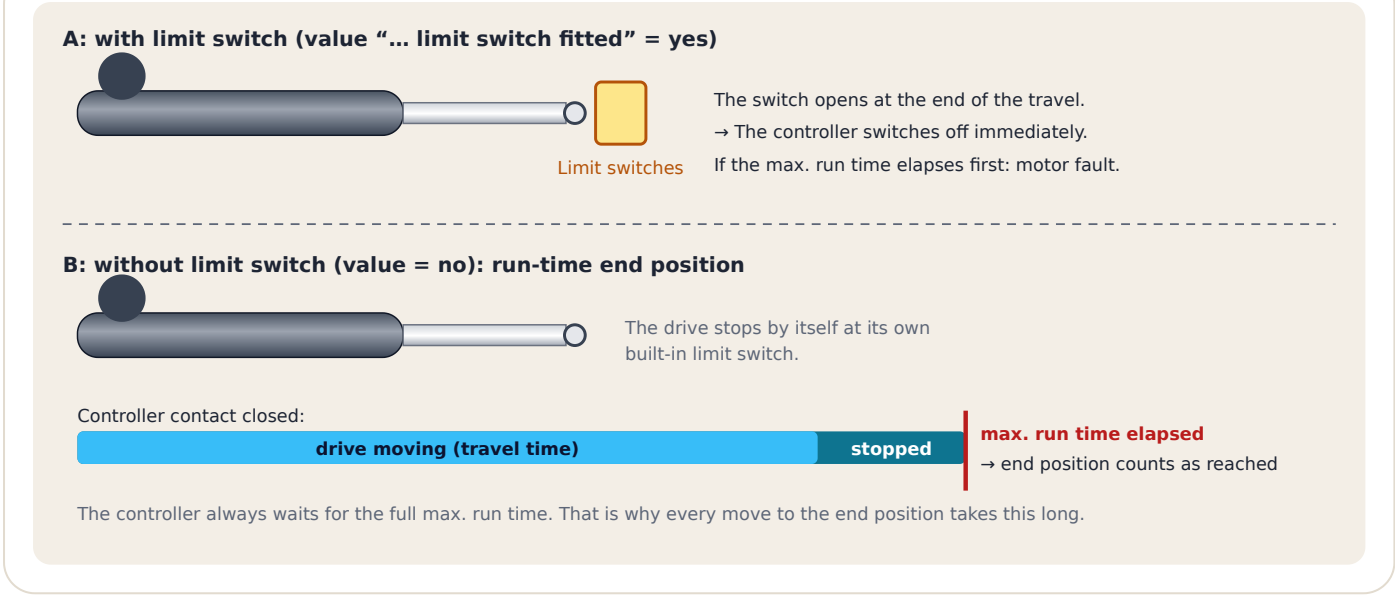


Fig. 74 – Limit switch or run-time end position: how the controller detects the end of the travel.

Elevation limit switch fitted

both `limitSwitchesElevation`

Default: **yes** Allowed: **yes / no** Applies: **only while drives are stopped**

Yes: limit switches (normally closed) for both ends of the tilt are connected to the circuit board. No: the drive stops at its own built-in limit switch; the controller detects the end by the max. run time.

SAMBA SOLAR RACK
no (no limit switches on the tracker).

OTHER RACK
Yes only if your own limit switches are connected to the terminals of the circuit board. If it is set to yes without limit switches, both directions are blocked (open input = end position).

Azimuth limit switch fitted

2 axes `limitSwitchesAzimuth`

Default: **yes** Allowed: **yes / no** Applies: **only while drives are stopped**

As above, for the rotation.

SAMBA SOLAR RACK
–

OTHER RACK
As above.

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Sun sensor EAST / WEST / SOUTH / NORTH on input AIN

both

`sunChEast, sunChWest, sunChSouth, sunChNorth`

Default: **EAST AIN0, WEST AIN1, SOUTH AIN2, NORTH AIN3**

Allowed: **AIN0 to AIN3, a different input for each side** Applies: **only while drives are stopped**

Which wire of the sun sensor is connected to which input. If a side is given an input that another side already has, the two swap.

SAMBA SOLAR RACK
Default (wire colours red = east, white = west, yellow = south, green = north according to the manufacturer).

OTHER RACK
Do not search by hand: the setup program (step 8) sets the assignment correctly itself while you shade the sensor.

North brighter means UP

both `sunNorthMeansUp`

Default: **yes** Allowed: **yes / no** Applies: **only while drives are stopped**

Determines in which direction the tilt moves when the NORTH side of the sensor is brighter than SOUTH. Depends on how the sensor is mounted (factory assumption: NORTH points to the top edge of the module).

SAMBA SOLAR RACK
Default; the test run checks it.

OTHER RACK
Do not change by hand: if the automatic moves away from the sun in the test run (step 12), the device switches the value itself.

Flat position is in UP direction

both `stormParkElevationPlus`

Default: **yes** Allowed: **yes / no**

Applies: **on the display immediately; remotely only after confirmation on the display**

In which direction (UP or DOWN contact) the rack becomes flat. Determines the move during storm, parking and night, and which terminal the STEEPER and FLATTER buttons switch.

SAMBA SOLAR RACK
Determine it on the rack with the setup program (step 5).

OTHER RACK
Determine it with the setup program (step 5): if you answer NO, the device switches the value itself.

Anemometer: baud rate

both `windBaud`

Default: **9600 baud** Allowed: **1200 to 115200**

Applies: **on the display immediately; remotely only after confirmation on the display**

Transmission speed on the RS485 line (8 data bits, no parity, 1 stop bit).

SAMBA SOLAR RACK
9600 (ACOVMELST factory setting).

OTHER RACK
From the data sheet of your anemometer.

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Anemometer: Modbus address both `windSpeedAddr`

Default: **3** Allowed: **1 to 247**

Applies: **on the display immediately; remotely only after confirmation on the display**

Address of the wind speed sensor on the bus.

SAMBA SOLAR RACK
3 (factory setting).

OTHER RACK
From the data sheet; each sensor on the bus needs a different address.

Anemometer: read function (3 or 4) both `windSpeedFunction`

Default: **3** Allowed: **3 or 4**

Applies: **on the display immediately; remotely only after confirmation on the display**

Modbus command for reading: 3 (holding register) or 4 (input register).

SAMBA SOLAR RACK
3.

OTHER RACK
From the data sheet (“function code 03” or “04”).

Anemometer: register both `windSpeedRegister`

Default: **0** Allowed: **0 to 65535**

Applies: **on the display immediately; remotely only after confirmation on the display**

Number of the register that holds the wind speed.

SAMBA SOLAR RACK
0.

OTHER RACK
From the data sheet (often given as 0x0000 = 0).

Anemometer: m/s per count both `windSpeedScale`

Default: **0.1** Allowed: **0.001 to 10**

Applies: **on the display immediately; remotely only after confirmation on the display**

Conversion of the raw value into m/s. 0.1 means: the sensor delivers tenths of m/s (raw value 86 = 8.6 m/s).

SAMBA SOLAR RACK
0.1.

OTHER RACK
From the data sheet. Check: with a known wind, the display must show roughly the same as a handheld anemometer.

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Wind direction: Modbus address, read function, register, ° per count

both `windDirAddr, windDirFunction, windDirRegister, windDirScale`

Default: **4 · 3 · 0 · 0.1** Allowed: **1–247 · 3/4 · 0–65535 · 0.001–10** Applies: **immediately**

The same details for the direction sensor (only with an anemometer with wind direction). 0.1 degree per count: raw value 0 to 3600 = 0° to 360°.

SAMBA SOLAR RACK
4 · 3 · 0 · 0.1 (ACOVMELST factory setting).

OTHER RACK
From the data sheet. The direction affects neither tracking nor storm protection.

Travel time 0 to 60° (steeper) / 60 to 0° (flatter) 1 axis

`elevTravelSteepMs, elevTravelFlatMs`

Default: **0 = not calibrated** Allowed: **0 or 10 to 300 s** Applies: **only while drives are stopped**

How long the tilt takes over the full travel. From this the device estimates every angle (tilt on command). Only valid if it is shorter than 0.9 × max. run time.

SAMBA SOLAR RACK
Measure it on your own rack with CALIBRATE TRAVEL TIME (setup program step 6).

OTHER RACK
Likewise with CALIBRATE TRAVEL TIME. Only enter by hand if the times are known exactly. Repeating it in autumn and spring is recommended (temperature, assumption).

Frame (tilt from stroke) 1 axis `elevGeometry`

Default: **linear** Allowed: **linear · G1 (2.0-D, 2.0-E) · G2 (2.0-E Plus, 2.1, 2.1 Plus)**

Applies: **only while drives are stopped**

How the travel of the drive is turned into an angle. G1 and G2 are the tables of the Samba Solar Racks (assembly manual, appendix 16.1).

SAMBA SOLAR RACK
G2 for 2.0-E Plus, 2.1, 2.1 Plus; **G1** for 2.0-D and 2.0-E.

OTHER RACK
linear. This is an assumption: with many frames, the angle does not change evenly with the stroke (on a G2 frame, linear would be off by up to about 2.6°). In the setup step “Move to angle” you can see how well it fits.

Start-up loss per move 1 axis `elevStartLossMs`

Default: **0 ms** Allowed: **0 to 3000 ms** Applies: **only while drives are stopped**

Time that each drive loses when starting and stopping. Improves the estimate with many short moves.

SAMBA SOLAR RACK
0 ms (not yet measured).

OTHER RACK
Keep 0 as long as it has not been measured.

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Hall tap present 1 axis `elevHall`

Default: **no** Allowed: **yes / no** Applies: **only while drives are stopped**

Yes: the HallTapBoard passes the pulses of the drive to the circuit board (input J22). The device then measures the tilt instead of estimating it, and detects whether the drive is really moving. With two axes it stays off.

SAMBA SOLAR RACK
yes if the HallTapBoard is fitted at the sync controller, otherwise no.

OTHER RACK
Yes only if a drive with a Hall sensor is connected to J22 via a HallTapBoard.

Invert Hall count direction 1 axis `elevHallInvert`

Default: **no** Allowed: **yes / no** Applies: **only while drives are stopped**

Reverses the counting direction if channels A and B are connected the other way round.

SAMBA SOLAR RACK
no; only change it if calibration reports “HALL COUNTS BACKWARDS”.

OTHER RACK
Likewise.

Hall steps per stroke 1 axis `elevHallSteps`

Default: **0 = not calibrated** Allowed: **0 or 1000 to 2000000** Applies: **only while drives are stopped**

How many Hall steps the drive counts over the full travel. Entered by calibration (mean of UP and DOWN).

SAMBA SOLAR RACK
Let calibration determine it.

OTHER RACK
Let calibration determine it; enter by hand only if the value is known.

Reset sync controller elevation both `syncResetElevation`

Default: **off** Allowed: **off · J23 · J5 (J5 only with 1 axis)** Applies: **only while drives are stopped**

Whether and where the reset input (RST) of a sync controller is connected. Some sync controllers lock after a power failure during a move until RST is briefly connected to COM; they then fully retract the drives.

SAMBA SOLAR RACK
J23 (BST-KZQ-15A-2 sync controller, circuit board from draft 0.12).

OTHER RACK
off if your controller has no reset input or it is not connected. J5 only with one axis, because the EAST contact is then free.

Reset sync controller azimuth 2 axes `syncResetAzimuth`

Default: **off** Allowed: **off · J23** Applies: **only while drives are stopped**

As above, for the rotation (terminal J23, pins 3–4).

SAMBA SOLAR RACK
–

OTHER RACK
As above.

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Location: latitude (north positive) / longitude (east positive) both

`siteLatitude, siteLongitude`

Default: **0 / 0 = not entered** Allowed: **latitude −90 to 90, longitude −180 to 180 degrees**

Applies: **immediately**

Where the rack stands. Needed for sunrise and sunset. Easiest via the map (Chapter 6.2).

SAMBA SOLAR RACK
Enter your location.

OTHER RACK
Enter your location.

Rack orientation (south 0, west positive) 1 axis `rackAzimuth`

Default: **0°** Allowed: **−90 to 90 degrees, whole degrees** Applies: **immediately**

Where the panels face (Chapter 6.3).

SAMBA SOLAR RACK
Measure it with a compass on the installed rack.

OTHER RACK
Likewise.

CONTROL

How the automatic tracks and when it is night. All values apply immediately.

Track from deviation both `trackStartPermille`

Default: **8 %** Allowed: **3 to 30 %** Applies: **immediately**

A tracking move only starts when the sun sensor sees more than this difference between two sides.

SAMBA SOLAR RACK
Default.

OTHER RACK
If the installation moves too often: set it higher. If it follows the sun too roughly: set it lower (more moves, more wear).

Aligned below deviation both `trackStopPermille`

Default: **2 %** Allowed: **0.5 to 10 %, always less than “Track from”** Applies: **immediately**

Below this difference the axis counts as aligned, and the move ends.

SAMBA SOLAR RACK
Default.

OTHER RACK
If an axis oscillates around the sun: set it a little higher.

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Delay before a move

both `trackStartDelayMs`

Default: **3 s** Allowed: **0 to 60 s** Applies: **immediately**

The deviation must last this long before a move starts. Prevents moves because of a passing cloud.

SAMBA SOLAR RACK
Default.

OTHER RACK
Set it longer with changing cloud cover.

Rest time after a move

both `trackSettleMs`

Default: **15 s** Allowed: **0 to 600 s** Applies: **immediately**

Pause after each tracking move.

SAMBA SOLAR RACK
Default.

OTHER RACK
Set it longer to reduce the number of moves.

Longest tracking move

both `trackMaxMoveMs`

Default: **20 s** Allowed: **1 to 120 s** Applies: **immediately**

A single tracking move ends after this time at the latest.

SAMBA SOLAR RACK
Default.

OTHER RACK
A little longer with very slow drives; always well below the max. run time.

Night below brightness (sum)

both `nightEnterLevel`

Default: **4000** Allowed: **0 to 100000** Applies: **immediately**

Only without time or location: if the sum of the four brightness values stays below this value long enough, it is night.

SAMBA SOLAR RACK
Default; at dusk, read the sum in the SUN SENSOR card and adjust.

OTHER RACK
Likewise.

Day above brightness (sum)

both `nightExitLevel`

Default: **6000** Allowed: **0 to 130000, always greater than the night threshold** Applies: **immediately**

Above this value it is day again. Also serves as the minimum brightness for the sun sensor and the test run in the setup program.

SAMBA SOLAR RACK
Default.

OTHER RACK
Likewise.

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Night or day only after both `nightDelayMs`

Default: **5 min** Allowed: **10 s to 60 min** Applies: **immediately**

The brightness must stay below or above the threshold for this long. Also applies to “cloudy”.

SAMBA SOLAR RACK
Default.

OTHER RACK
Likewise.

Return east at night 2 axes `nightReturnEast`

Default: **yes** Allowed: **yes / no** Applies: **immediately**

Yes: at night, after the flat move, the azimuth turns to the east so that the installation is ready in the morning.

SAMBA SOLAR RACK
–

OTHER RACK
Keep yes, unless the mechanics should not turn at night.

Night from sunset plus both `nightAfterSunsetMin`

Default: **30 min** Allowed: **0 to 180 min** Applies: **immediately**

With time and location: night begins this many minutes after sunset.

SAMBA SOLAR RACK
Default.

OTHER RACK
Likewise.

Day from sunrise minus both `dayBeforeSunriseMin`

Default: **0 min** Allowed: **0 to 180 min** Applies: **immediately**

Night ends this many minutes before sunrise.

SAMBA SOLAR RACK
Default.

OTHER RACK
Likewise.

When cloudy, move to calculated angle 1 axis `cloudAstro`

Default: **no** Allowed: **yes / no** Applies: **immediately**

When cloudy, the tilt moves to the calculated best angle (Chapter 6.5).

SAMBA SOLAR RACK
As you wish; requires time, location and a calibrated travel time.

OTHER RACK
Likewise.

Cloudy below brightness (sum) 1 axis `cloudLevel`

Default: **20000** Allowed: **0 to 130000** Applies: **immediately**

Below this sum it counts as cloudy (assumption, set it on the device).

SAMBA SOLAR RACK
Read it on an overcast day.

OTHER RACK
Likewise.

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Preset angle 1 / preset angle 2 1 axis `elevPreset1, elevPreset2`

Default: **0 = off** Allowed: **0 or 1 to 59°, not 30, not twice** Applies: **immediately**

Up to two angles of your own as additional buttons in the TILT card (next to 0°, 30°, 60°).

SAMBA SOLAR RACK
As you wish, for example an angle for winter.

OTHER RACK
Likewise.

DIAGNOSTICS

For troubleshooting only.

Log level (0 off, 5 all) both `logLevel`

Default: **4** Allowed: **0 to 5** Applies: **immediately, until the restart (not saved)**

How detailed the device's log is (0 off, 1 errors, 2 warnings, 3 info, 4 debug, 5 all).

SAMBA SOLAR RACK
Default.

OTHER RACK
Only change it for troubleshooting.

Fixed values you cannot set

Some protective values are fixed in the firmware. You should know them, because they explain why the rack sometimes waits.

PROTECTION	VALUE	WHAT YOU NOTICE
Duty cycle: continuous move	at most 120 s	When a drive's travel time account is empty, a pause begins. The status bar shows in orange “DUTY CYCLE PAUSE: DRIVE AVAILABLE AGAIN IN mm:ss”. After a complete emptying it lasts about 13.5 min. Only the flat move during storm, anemometer failure and parking still runs.
Duty cycle: share	10 %; available again from 90 s saved in the account (assumption)	
Repeated flat move during storm (without limit switches)	every 30 min	During storm and anemometer failure, the elevation regularly moves flat again in case an earlier move had no effect.
Test move in the setup program	at most 5 s	The FLATTER or EAST test move stops by itself.

The duty cycle values are dealer specifications for the drives of the Samba Solar Rack (10 % duty cycle, 120 s continuous). Other drives may tolerate more or less; this cannot be set in firmware 0.7.0.

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8 Web interface: settings from a distance

When Wi-Fi is connected, you can view, operate and set up the installation in the browser of a phone or PC. The page looks like the display. The display remains the main device, though: everything available remotely is also available on the display, and some things are only available there.

> 🧩 **Testing status**
> Viewing and updating over the network have been tried out on the test board (in simulation, without circuit board). Operating and setting remotely have not yet been tried out there; location map, sun and test report are new in 0.7.0 and only checked on the PC.

Opening and logging in

1 Open the address

In the same Wi-Fi, type the address from the NETWORK screen into the browser, or `http://samba-solar-track.local`. *Result:* the overview appears; without logging in, “VIEW ONLY” is shown at the top.

2 Log in

On a phone, scan the QR code under NETWORK → REMOTE ACCESS. On a PC, click ENTER TOKEN and type in the four blocks of the token shown on the NETWORK screen. *Result:* the browser may operate and change settings; it remembers the login.

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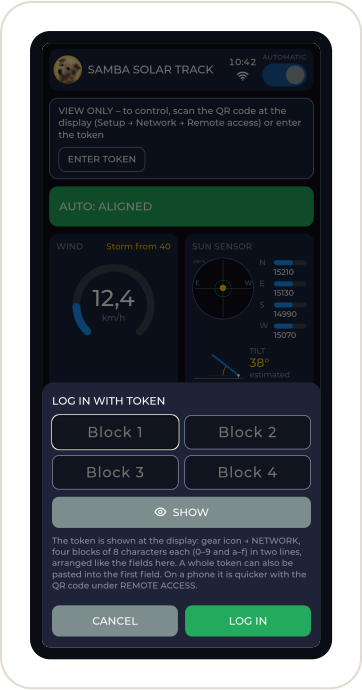


Fig. 75 – Logging in on a PC: the token in four blocks.



Fig. 76 – “Remote control locked at the display”: view only.

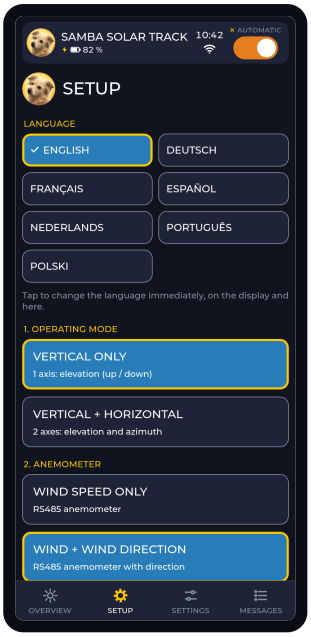


Fig. 77 – SETUP in the browser: language, operating mode, anemometer.



Fig. 78 – 1 axis Overview on a PC, with the TILT card.



Fig. 79 – 2 axes Overview on a PC with EAST and WEST.

What can be done remotely

PAGE	SETTING	HOW IT TAKES EFFECT
SETUP	Language, design	immediately, on the display and in the browser; only while the dashboard is open on the display
SETUP	Operating mode, anemometer	like SAVE on the display: drives stop, restart after 1.5 s
SETUP	Location (map), rack orientation 1 axis	immediately
SETUP	Setup program: status and test reports (each VIEW, save as PDF, and TEXT)	view, save and download only

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PAGE	SETTING	HOW IT TAKES EFFECT
SETUP	Upload firmware or install it from the internet; end simulation	with token; not during storm and not while a drive is moving
SETTINGS	all values from Chapter 7	Control and commissioning immediately or at standstill; safety values, direction of the flat position and wind speed data only after confirmation on the display
SETTINGS	RESET SYNC CONTROLLER	with confirmation prompt in the browser

Location and orientation in the browser

In the browser there is a street map (Leaflet with maps from OpenStreetMap) and a place search (Nominatim). Your browser loads the map images from the internet, not the device; without internet, the number fields still work.

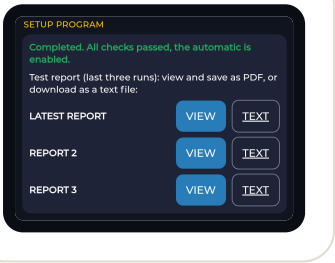


Fig. 80 – Detail: status of the setup program with the test reports, each with VIEW and TEXT.

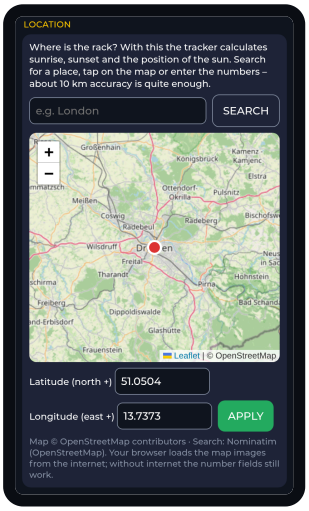


Fig. 81 – Detail: map with red dot, latitude and longitude.



Fig. 81a – 1 axis Detail: RACK ORIENTATION.

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Fig. 82 – 1 axis Below the overview, the SUN section with the sun's daily arc.



Fig. 83 – As long as the setup program is missing: violet bar “CONTINUE AT THE DISPLAY”.

What only works on the display, and why

ONLY ON THE DISPLAY	WHY
Going through the setup program	During every test move, someone must stand at the rack and watch. The browser only shows the progress.
CALIBRATE TRAVEL TIME 1 axis	Same reason: you tap ARRIVED when the frame stops. During calibration, the device rejects manual moves from remote.
Confirming safety values	A value that protects the mechanics should not be changed remotely without anyone noticing.
Wi-Fi name and password	Without Wi-Fi there is no browser. Alternatively via the setup hotspot (4.4).
QR code, show token, RENEW TOKEN, lock REMOTE CONTROL	Whoever stands at the device decides who may control it remotely.
Switching on the simulation	Needs a confirmation on the device and only works without circuit board.
Sound, screen (idle mode, motion sensor, battery)	Only concerns the device itself.

ℹ️ **The REMOTE CONTROL switch locks everything**

When locked, the device accepts no command, no change and no firmware update from remote. Displays and messages continue. Storm protection, limit switches, dead time and run time monitoring can never be overridden remotely.

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9 Troubleshooting the setup

The most frequent messages and observations during setup, with cause and remedy. Wiring faults are covered by the wiring manual (German only), circuit board faults by the Assembly and Operation Manual.

DISPLAY OR OBSERVATION	CAUSE	REMEDY
Everything in English	The default language is English.	In the language selection, tap the desired language, for German DEUTSCH (4.1).
SAVE + RESTART cannot be tapped	Operating mode or anemometer not selected.	Select both.
SETTINGS button missing	Setup not saved yet.	First SAVE + RESTART.
READY flashes, nothing moves	Setup missing.	Save the setup.
Violet bar “SETUP PROGRAM MISSING: AUTOMATIC LOCKED”	Setup program not completed.	Tap the bar and continue (Chapter 5).
After plugging on the circuit board: violet bar “SETUP PROGRAM MISSING”, although the setup program was finished	It was only completed in the simulation. On the real rack it then starts again by itself.	Intended. Go through the setup program on the rack (Chapter 5).
“SIMULATION NOT POSSIBLE”	The RearCircuitBoard_10 board is plugged on.	With the circuit board there is no simulation; this is intended.
Step 3: field 0x70 red	Circuit board not plugged on properly, chip faulty.	Supply off, check the circuit board on the socket header J8. Without 0x70 nothing moves.
Step 3: 24 V or 5 V red	Controller power supply off, fuse, converter.	Check power supply and fuse (wiring manual, German only).
Step 3: BATT red	Charge lock implausible.	Assembly and Operation Manual, Chapter 7.
Step 3: 0x72 red	Reset at J23 selected, chip does not respond (circuit board before draft 0.12?).	In step 2, set reset to “off” or check the circuit board.
“The wind wheel has not been turned yet.”	Reading rose by less than 2 km/h, or the anemometer does not respond.	Turn it harder. If it does not respond: RS485 line (A/B swapped?), supply, address and baud rate.
Wind direction does not show north	Sensor mounted twisted or wrong values.	Align the sensor's north mark, check the wind direction values, REPEAT.
With FLATTER the rack became steeper	Direction of the flat position the other way round.	Tap NO: the device switches it. REPEAT.

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DISPLAY OR OBSERVATION	CAUSE	REMEDY
Nothing moves with FLATTER	Drive power supply off, contacts not connected, sync controller not in jog mode or locked (for example after a power failure).	Check the wiring; with a sync controller with reset: set up and trigger the reset.
“Drive fault (maximum run time). Acknowledge and repeat.”	With “limit switch fitted = yes”, the limit switch did not trip before the max. run time elapsed.	Limit switch present? Otherwise set it to no. Otherwise increase the max. run time. ACKNOWLEDGE, REPEAT.
Both directions blocked, nothing moves	“Limit switch fitted = yes”, but no limit switches connected (open input = end position).	Set it to no (limit switch).
“Drive stopped without end position (Hall pickup).”	Hall pulses stopped in the middle of the travel: mechanics jammed, drive without voltage, sync controller aborted.	Check the mechanics and the drive supply, REPEAT.
“The travel time has not been calibrated yet.”	CALIBRATE TRAVEL TIME not completed.	Calibrate and APPLY.
Calibration: “MAX. RUN TIME TOO SHORT OR ARRIVED NOT TAPPED”	Move took longer than the max. run time, or ARRIVED forgotten.	Increase the max. run time, measure again.
Calibration: “HALL COUNTS BACKWARDS”	Channels A and B swapped.	Invert Hall count direction = yes.
With EAST the rack turns west	EAST and WEST contacts swapped.	Power supply off, swap the plugs at J5 and J6, REPEAT.
“Reset not possible (check connection or chip).”	Reset did not start within 20 s.	Check the terminal and the reset setting.
“Too dark: … repeat.”	Brightness below the day threshold.	In daylight or with a bright lamp in front of the sensor.
Sun sensor: side not detected	Shadow too small (less than 25 % darker) or only brief.	Hand closer, shadow over the whole side, hold for 1 s.
“No time yet: set up WiFi or skip.”	Wi-Fi off or time server not reachable.	Set up Wi-Fi (4.4) or SKIP.
Time wrong by one or more hours	Wrong time zone.	NETWORK → TIME → SUGGESTIONS.
“No location yet: enter it or skip.”	Latitude and longitude are 0.	Enter the location (6.2).
Sunrise and sunset do not match your location	Latitude and longitude swapped or wrong sign.	Latitude = north, longitude = east; check.
Test run: “The direction has been switched. Please once more.”	Automatic moved away from the sun.	REPEAT; the device has already corrected the value.
“The automatic mode did not align. Is there enough sun?”	Not aligned after 3 min.	Repeat in sunshine; sensor free of shadow?

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DISPLAY OR OBSERVATION	CAUSE	REMEDY
“Storm: the rack moves flat. Repeat afterwards.”	Gust above the storm threshold.	Wait for the end of the storm and the hold time, REPEAT.
Orange “DUTY CYCLE PAUSE …”	Drive's travel time account empty.	Wait; the remaining time is shown (Chapter 7).
Value is rejected	Outside the limits, does not fit another value, or a drive is moving.	Read the reason; with “only while drives are stopped”, wait a moment.
Browser: change is not accepted	Not logged in, REMOTE CONTROL locked, or the dashboard is not open on the display (language, design).	Log in, release the lock on the display, return the display to the dashboard.

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10 Terms simply explained

Axis	A direction in which the rack can move. Elevation = tilt, azimuth = rotation.
Automatic	The controller tracks the rack to the sun by itself, moves to the night position at night and back in the morning. AUTOMATIC switch in the header.
Azimuth	The east-to-west rotation, only with 2 axes. Also: the compass direction of the sun in degrees.
Gust	Here: the average wind speed over 3 seconds. If it is above the storm threshold, it counts as a storm.
Decimal degrees	Notation for latitude and longitude with decimal places, for example 51.0504 instead of 51° 3′ 1″.
Calibration	The device measures the travel time (and, with a Hall tap, the Hall steps) over the full travel of the tilt. Only with 1 axis.
Duty cycle	How long a drive may run in relation to the pause without overheating. Here 10 % and at most 120 s continuously.
Setup	The screen with operating mode and anemometer. Without a saved setup, no drive moves.
Elevation	The tilt of the module between flat and steep.
End position	The end of the travel in one direction.
Limit switches	A switch at the end of the travel that opens when reached (normally closed, NC). For the controller, open means: direction blocked.
Flat position	The flat position of the module. Also the storm position and night position (0 position).
Frame G1, G2	The designs of the Samba Solar Racks. They determine how the travel of the drive is turned into an angle.
Hall tap, Hall steps	Many drives have a Hall sensor that gives pulses while moving. The HallTapBoard passes them on to the circuit board. A Hall step is one counted edge of these pulses.
Hold time	How long the rack stays flat after the last storm gust.
Manual mode	Automatic off: the rack only moves while you hold a move button.

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Max. run time	Longest permitted continuous move. Without limit switches, the controller uses it to detect the end position.
Hotspot	A separate, temporary Wi-Fi network of the device for entering the Wi-Fi details with a phone.
Run-time end position	Without limit switches: after the full max. run time in one go, the end position counts as reached.
Modbus, RS485	The way the anemometer talks to the circuit board over two wires. Each sensor on the bus has an address.
NTP, time server	A service on the internet from which the device fetches the exact time.
0 position	The night position: elevation flat.
Test run	Last test step of the setup program: the automatic must turn the offset rack back to the sun.
Test report	Report with all results and measured values of the setup program, in the web interface as a designed sheet (PDF) or as a text file; the last three are kept in the device.
Reset of the sync controller	A short contact between RST and COM with which some sync controllers release themselves again after a fault; in doing so they fully retract the drives.
Setup program	The guided check of the whole installation after the setup. Only then is the automatic enabled.
Simulation	Operation without circuit board with a simulated rack, only for trying things out.
Sun sensor	A sensor with four sides (east, west, south, north). From the difference in brightness the controller detects where the sun is.
Storm threshold	Wind speed from which the rack moves flat.
Sync controller	A control unit that makes the two drives of an axis move at exactly the same speed (for example BST-KZQ-15A-2). The circuit board “presses” its buttons.
Token	Secret key of 32 characters. Whoever has it may operate the device remotely.
Dead time	Pause before a change of direction.
Web interface	The control page in the browser of a phone or PC, in the home Wi-Fi.

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11 Appendix

A. All values at a glance

Applies: “immediately”, “standstill” = only while drives are stopped, “confirmation¹” = on the display immediately, remotely only after confirmation on the display. Axes: 1 = only 1 axis, 2 = only 2 axes, 1+2 = both. Samba Solar Rack column: short form, details in Chapter 7. Details and recommendations for other racks: click the names.

VALUE (DISPLAY)	AXES	DEFAULT	PERMITTED	APPLIES	SAMBA SOLAR RACK
SAFETY					
Storm threshold (gust)	1+2	40 km/h	20 to 60 km/h	confirmation¹	40 km/h
Hold time after storm	1+2	10 min	5 to 60 min	confirmation¹	10 min
Dead time on direction change	1+2	500 ms	300 to 5000 ms	confirmation¹	500 ms
Max. run time elevation	1+2	60 s	5 to 300 s	confirmation¹	90 s
Max. run time azimuth	2	90 s	5 to 300 s	confirmation¹	–
COMMISSIONING					
Elevation limit switch fitted	1+2	yes	yes / no	standstill	no
Azimuth limit switch fitted	2	yes	yes / no	standstill	–
Sun sensor EAST / WEST / SOUTH / NORTH on input AIN	1+2	EAST AIN0, WEST AIN1, SOUTH AIN2, NORTH AIN3	AIN0 to AIN3, a different input for each side	standstill	Setup program
North brighter means UP	1+2	yes	yes / no	standstill	Setup program
Flat position is in UP direction	1+2	yes	yes / no	confirmation¹	Setup program
Anemometer: baud rate	1+2	9600 baud	1200 to 115200	confirmation¹	9600
Anemometer: Modbus address	1+2	3	1 to 247	confirmation¹	3
Anemometer: read function (3 or 4)	1+2	3	3 or 4	confirmation¹	3
Anemometer: register	1+2	0	0 to 65535	confirmation¹	0
Anemometer: m/s per count	1+2	0,1	0.001 to 10	confirmation¹	0,1
Wind direction: Modbus address, read function, register, ° per count	1+2	4 · 3 · 0 · 0,1	1–247 · 3/4 · 0–65535 · 0,001–10	immediately	4 · 3 · 0 · 0,1
Travel time 0 to 60° (steeper) / 60 to 0° (flatter)	1	0 = not calibrated	0 or 10 to 300 s	standstill	calibrate

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VALUE (DISPLAY)	AXES	DEFAULT	PERMITTED	APPLIES	SAMBA SOLAR RACK
Frame (tilt from stroke)	1	linear	linear · G1 (2.0-D, 2.0-E) · G2 (2.0-E Plus, 2.1, 2.1 Plus)	standstill	G2 (2.0-E Plus)
Start-up loss per move	1	0 ms	0 to 3000 ms	standstill	0 ms
Hall tap present	1	no	yes / no	standstill	yes with HallTapBoard
Invert Hall count direction	1	no	yes / no	standstill	no
Hall steps per stroke	1	0 = not calibrated	0 or 1000 to 2000000	standstill	calibrate
Reset sync controller elevation	1+2	off	off · J23 · J5 (J5 only with 1 axis)	standstill	J23
Reset sync controller azimuth	2	off	off · J23	standstill	–
Location: latitude (north positive) / longitude (east positive)	1+2	0 / 0 = not entered	latitude −90 to 90, longitude −180 to 180 degrees	immediately	your location
Rack orientation (south 0, west positive)	1	0°	−90 to 90 degrees, whole degrees	immediately	measure
CONTROL					
Track from deviation	1+2	8 %	3 to 30 %	immediately	Default
Aligned below deviation	1+2	2 %	0.5 to 10 %, always less than “Track from”	immediately	Default
Delay before a move	1+2	3 s	0 to 60 s	immediately	Default
Rest time after a move	1+2	15 s	0 to 600 s	immediately	Default
Longest tracking move	1+2	20 s	1 to 120 s	immediately	Default
Night below brightness (sum)	1+2	4000	0 to 100000	immediately	Default
Day above brightness (sum)	1+2	6000	0 to 130000, always greater than the night threshold	immediately	Default
Night or day only after	1+2	5 min	10 s to 60 min	immediately	Default
Return east at night	2	yes	yes / no	immediately	–
Night from sunset plus	1+2	30 min	0 to 180 min	immediately	Default
Day from sunrise minus	1+2	0 min	0 to 180 min	immediately	Default
When cloudy, move to calculated angle	1	no	yes / no	immediately	as you wish

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VALUE (DISPLAY)	AXES	DEFAULT	PERMITTED	APPLIES	SAMBA SOLAR RACK
Cloudy below brightness (sum)	1	20000	0 to 130000	immediately	as you wish
Preset angle 1 / preset angle 2	1	0 = off	0 or 1 to 59°, not 30, not twice	immediately	as you wish
DIAGNOSTICS					
Log level (0 off, 5 all)	1+2	4	0 to 5	until restart	Default

B. My values (for printing)

Enter here what you have set and measured on your installation. The measured values of the setup program are also in the test report.

ITEM	MY VALUE	DATE
Operating mode (1 or 2 axes), anemometer		
Limit switch elevation / azimuth		
Max. run time elevation / azimuth		
Measured travel time steeper / flatter		
Travel time east / west / back to east (2 axes)		
Hall steps per stroke		
Angle read at 30° (1 axis)		
Flat position in UP direction (yes/no)		
Location latitude / longitude		
Rack orientation (1 axis)		
Brightness (sum) at dusk / when cloudy		
Device name, address in the Wi-Fi		

C. Assumptions and open points

ITEM	STATUS
Firmware 0.7.0 with setup program, sun position and location	only tested on the PC and in simulation, not on the rack
Limits of the setup program (travel times, Hall steps, angles)	none; will be set on the rack after delivery of the drives (E-S3)
0 position = flat position (storm direction)	assumption for all drive variants
Mounting position of the sun sensor (NORTH towards the top edge)	assumption; the setup program corrects the direction
Threshold “cloudy” 20000	assumption, set it on the device

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ITEM	STATUS
Reset of the sync controller: pulse 5 s, behaviour afterwards	assumption, measure on the rack
Release of the duty cycle after 90 s saved in the account	assumption, check on the rack
Max. run time 90 s and frame G2 for the Rack 2.0-E Plus	from the README; check against the current assembly manual
Frame “linear” for other racks	assumption

D. Further documents

- Assembly and Operation Manual (`index.html`): circuit board, sensors, firmware, operation.
- Wiring manual (`verkabelung.html`, German only): all cables.
- Assembly manual of the Samba Solar Rack: website and shop.

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12 Licence and copyright

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Sources of maps and places

WHAT	SOURCE	LICENCE
Outlines of countries and coasts in the map on the display	Natural Earth	public domain
Place names for the search on the display	GeoNames	CC BY 4.0
Map display in the web interface	Leaflet 1.9.4 (included in the device)	BSD-2-Clause
Map images and place search in the web interface	OpenStreetMap (tiles), Nominatim; loaded by the browser from the internet	© OpenStreetMap contributors, ODbL

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The documents are provided without warranty. Building, setup and operation are at your own risk.

13 Change history

What has changed in this manual, newest change first.

EDITION	DATE	CHANGE
0.7.6	11/2026	Test report: in the web interface VIEW as a designed sheet (light or dark) and SAVE AS PDF, next to it TEXT; picture of the FIRMWARE page with 0.7.6.
0.7.0	11/2026	First edition for firmware 0.7.0: setup on the display, setup program for one and two axes, location, orientation, sun and night, all adjustable values with recommendations for the Samba Solar Rack and other racks, web interface, troubleshooting. Firmware only tested on the PC.

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First edition, November 2026, firmware 0.7.0. Accompanies the Assembly and Operation Manual and the wiring manual (German only).